

Data & Analytics

Data Extraction Tool for SAP® Execution Guide

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1. Overview

1.1. Introduction

This document describes how to extract data from SAP using the Data Extraction Tool using Advanced Business Application Programming (ABAP).

The purpose of the Data Extraction Tool is to extract the necessary data required to run Data & Analytics procedures require the extraction of data from some of the largest transaction data tables in SAP.

This extraction is complicated and needs a table specific approach. The tool designed the logic in the ABAP program to avoid potential threats such as:

- Long running extraction statements against the SAP Database Management System (DBMS) which may cause the program to stop running.
- Extracting unnecessary data due to improper selection criteria.
- Overloading the available memory (RAM) during the data selection process which may impact the system performance or other background processes.

There are two program guides that describe the Data Extraction Tool in detail. The Execution Guide (this document) is written for anyone who uses the Data Extraction Tool. The Implementation Guide provides a step-by-step overview of how to implement ABAP programs.

1.2. Data Extraction Tool

The Data Extraction Tool for SAP contains Read Only statements which produce output files that are then sent to a user-specified directory in the SAP application server. The 2020 Data Extraction Tool can only be used in SAP release version S/4HANA, ECC 6.0 environment.

The extraction selections are defined as agreed to during the audit scoping process. Once the implementation of the Data Extraction Tool is complete, they are ready to extract the relevant data.

There are two types of SAP data that are extracted

- The Screening data are information about the organizational structure, the financial account structure as well as basic system settings for planning purposes.
- The Transaction data are the actual business transaction and master data from the SAP system. Depending on the scope of the Audit, only a subset of these data might need to get executed.

1.3. Extraction process

The following diagram describes the steps needed to implement the Data Extraction Tool and run the extraction in the production environment.

Data Extraction Tool and



Before transporting the reports to production, test them in a test environment.

As soon as the reports are available in the production environment perform the following steps:

- import/upload the scoping and configuration data
- extract the Screening data
- ZIP the output files
- encrypt the files
- transfer the files to

The results of the Screening data extraction must be validated by the transaction data are extracted from production. Once the Screening data are valid will contact you to discuss the de when to extract the transaction data. This includ relevant SAP periods for the extraction. will send a file that contains the scoping parameters and configuration data for the transactio xtraction. You can start the transaction data extraction in production after the last relevant SAP period is closed for postings.

When the transaction data extraction process is complete, the transaction data files are transferred to via s the secure file transfer address with the instructions on how to use sFTP is d by A secondary option of data transfer may be encrypted external hard disk. A detailed ov n how to execute the Screening and transaction data extraction can be found in Section II of this document.

1.4. Before you begin

The following prerequisites must be met prior to executing the ABAP report:

- Create an extraction folder with sufficient disk space on the SAP application server to store the extracted files. Central Team will assist you with the calculation of the required data with the 'File Size estimation feature' in the Tool.
- Verify that the user executing the ABAP reports has sufficient write permissions on the extraction folder/directory. The user must also have specific SAP authorizations (Table read permissions via object S_TABU_DIS or S_TABU_NAM). This is an additional security measure to restrict regular users with **SA38** permission from executing these ABAP reports. The user must also have SAP authorization to upload data (S_GUI ACTVT = 60) to be able to upload the XML file containing the scope and configuration data.

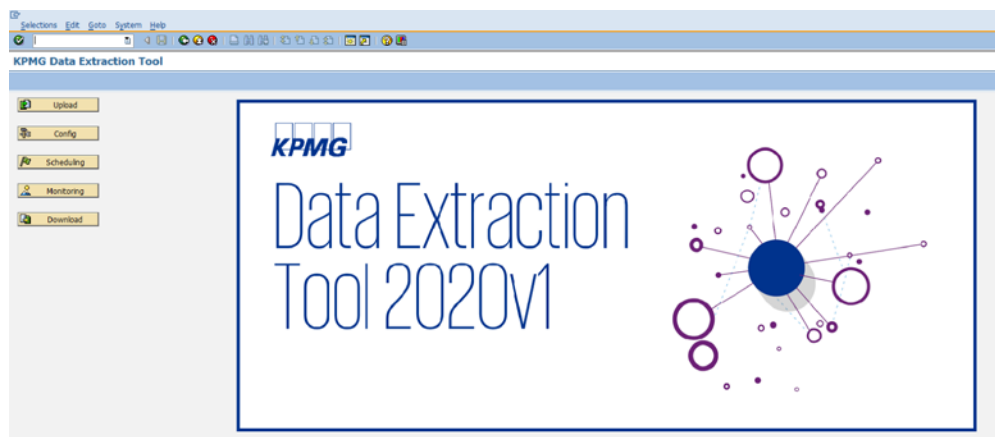
2. Execution of the Data Extraction Tool

2.1. Importing Scope and Configuration Data

Go to transaction SE38 or SA38 and execute the program A/DATA_EX_TOOL_20 to view Data Extraction Tool's **landing page**.

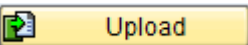
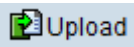
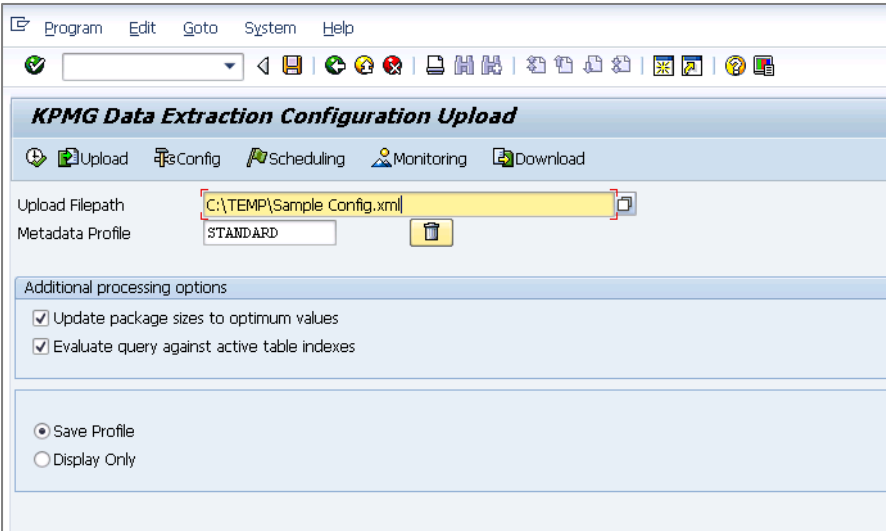
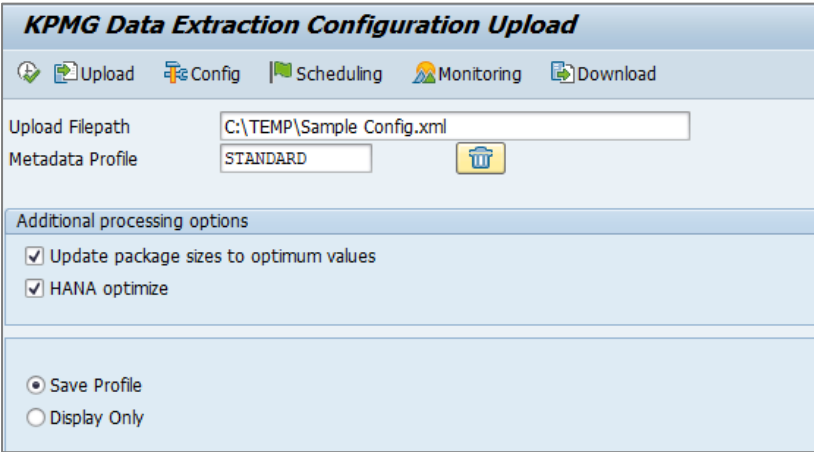
You can also execute the transaction A/DATA_EX_20 to view this page.

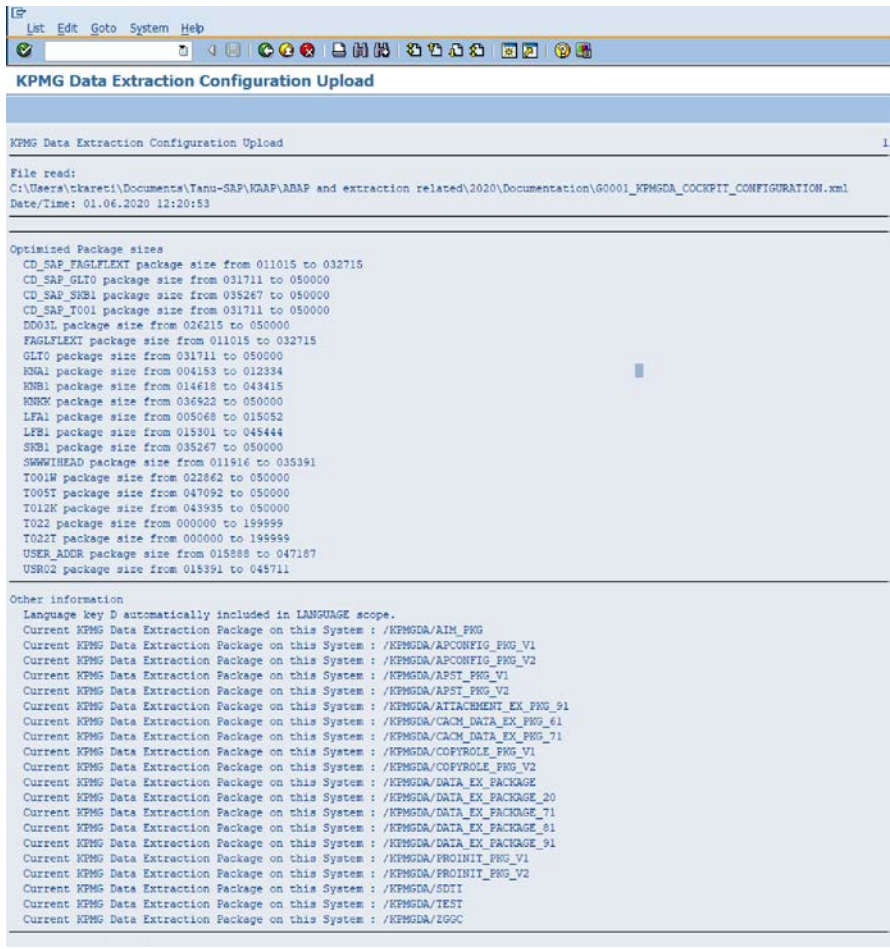
This page provides the steps of the data extraction workflow.



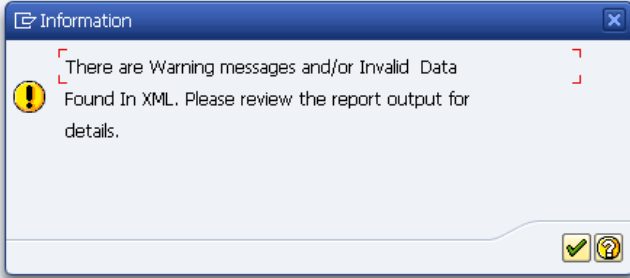
In order to execute data extraction with the Data Extraction Tool, a **Metadata Profile** is required. This profile fills the **Constraints** fields and provides the configuration of the **Table Groups**, **Technical**, **Destination**- and **Source Settings** in the selection screen.

In the figure below, the Metadata Profile **STANDARD** contains the scoping parameters to fill up the defaults for: **Client**, **Fiscal Year**, **Company Code**, **Object classes**, **Period**, and **Language** fields. It also contains the table groups and the associated SAP tables, the table fields to be extracted, their options or constraints for data selection which fill up the selection screen block labeled **Table Groups**. Other configuration data such as **Extraction Logic**, **Parallel Processing option**, **Packages sizes**, etc., are used to fill up the **Technical Settings** block.

Step	Action
	From the landing page, click  From the different other pages, click  Result: Upload selection screen displays This screen varies for SAP ECC 6.0 version and SAP S/4 HANA version of the tool. Below is the Upload selection screen for SAP ECC 6.0 version.  Below is the Upload selection screen for SAP S/4 HANA version 
2.	– Enter desired Metadata Profile field. It is recommended to use a new Metadata Profile name when uploading a new XML file.(Hint: To view all the existing variants, press F4 on the field to browse all existing variant names) – Enter name of file to import in Upload Filepath field. (Hint: press F4 on the field to browse and search for the file.) – Select Save Profile using Program and Variant names to save and store the Metadata Profile. – Select Update package sizes to optimum values to adjust the default package size values in the XML file to suit the memory requirements of the client's system. Package size is number of records to be extracted in one

Step	Action
	iteration from the database. For SAP ECC 6.0 version - Select Evaluate query against active table indexes to adjust the extraction logic of certain tables as per the indexes active on the client's system for faster data extraction. For SAP S/4 HANA version - Select the option 'HANA optimize'. This option switches the extraction logic of all tables to flat query selection method. - Languages referenced by the company codes in scope, are inserted in the Metadata profile (if they are not in the XML file). - Select Display Only when you only want to check the contents of the file without saving. (If you only want to display the contents of a saved Metadata Profile, enter the Metadata Profile ID, remove the Upload Filepath and then select Display only.)
3.	Click  Result: The file is read and the Metadata Variant is saved if the checkbox was checked. A report of the file contents is displayed.  The screenshot shows a window titled "KPMG Data Extraction Configuration Upload". It displays the file path: "C:\Users\ckareti\Documents\Tanu-SAP\KAP\ABAP and extraction related\2020\Documentation\G0001_KPMGDA_COCKPIT_CONFIGURATION.xml" and the date/time: "01.06.2020 12:20:53". Below this, it lists "Optimized Package sizes" for various SAP packages like CD_SAP_FAGLFLEX, CD_SAP_GLTO, etc., along with their size ranges. At the bottom, it lists "Other information" including the language key 'D' and a list of current KPMG Data Extraction Packages on the system, such as /KPMGDA/ADM_PKG, /KPMGDA/APCONFIG_PKG_V1, etc.

Step	Action
	The tables and field names in the configuration file are verified against the data dictionary and if invalid names are found, they will be listed in the Error list at the beginning of the report. <pre>List Of Errors In XML Invalid: <Field> MANDT33 in <Table> EKUB Invalid: <Field> MATNR44 in <Table> EKUB List Of Errors In XML Invalid: <Field> MANDT33 in <Table> EKUB Invalid: <Field> MATNR44 in <Table> EKUB</pre> If package sizes are optimized the changes will be listed in the report output <pre>Optimized Package sizes CD_SAP_FAGLFLEXT package size from 011015 to 032715 CD_SAP_GLTO package size from 031711 to 050000 CD_SAP_SKB1 package size from 035267 to 050000 CD_SAP_T001 package size from 031711 to 050000 DD03L package size from 026215 to 050000 FAGLFLEXT package size from 011015 to 032715 GLTO package size from 031711 to 050000 KNAL package size from 004153 to 012334</pre> If any changes are made to the extraction logic or constraints for optimizing data extraction, then relevant message is Upload program report. <pre>Adjusted extraction logic BKPF GJAHR removed from SQL CDHDR switched to DaybyDay Date-delimited by UDATE LIKP switched to DaybyDay Date-delimited by LFDAT VBAK switched to DaybyDay Date-delimited by AUDAT VBRK switched to DaybyDay Date-delimited by FKDAT</pre> All the current Data Extraction Packages are also displayed in this report. <pre>Other information Language key D automatically included in LANGUAGE scope. Current KPMG Data Extraction Package on this System : /KPMGDA/APCONFIG_PKG_V1 Current KPMG Data Extraction Package on this System : /KPMGDA/APCONFIG_PKG_V2 Current KPMG Data Extraction Package on this System : /KPMGDA/APST_PKG_V1 Current KPMG Data Extraction Package on this System : /KPMGDA/APST_PKG_V2 Current KPMG Data Extraction Package on this System : /KPMGDA/ATTACHMENT_EX_PKG_91 Current KPMG Data Extraction Package on this System : /KPMGDA/CACM_DATA_EX_PKG_61 Current KPMG Data Extraction Package on this System : /KPMGDA/CACM_DATA_EX_PKG_71 Current KPMG Data Extraction Package on this System : /KPMGDA/COPYROLE_PKG_V1 Current KPMG Data Extraction Package on this System : /KPMGDA/COPYROLE_PKG_V2 Current KPMG Data Extraction Package on this System : /KPMGDA/DATA_EX_PACKAGE Current KPMG Data Extraction Package on this System : /KPMGDA/DATA_EX_PACKAGE_20 Current KPMG Data Extraction Package on this System : /KPMGDA/DATA_EX_PACKAGE_71 Current KPMG Data Extraction Package on this System : /KPMGDA/DATA_EX_PACKAGE_81 Current KPMG Data Extraction Package on this System : /KPMGDA/DATA_EX_PACKAGE_91</pre> A warning message is displayed to prompt that there are some exception messages that need to be verified.

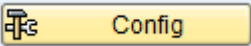
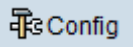
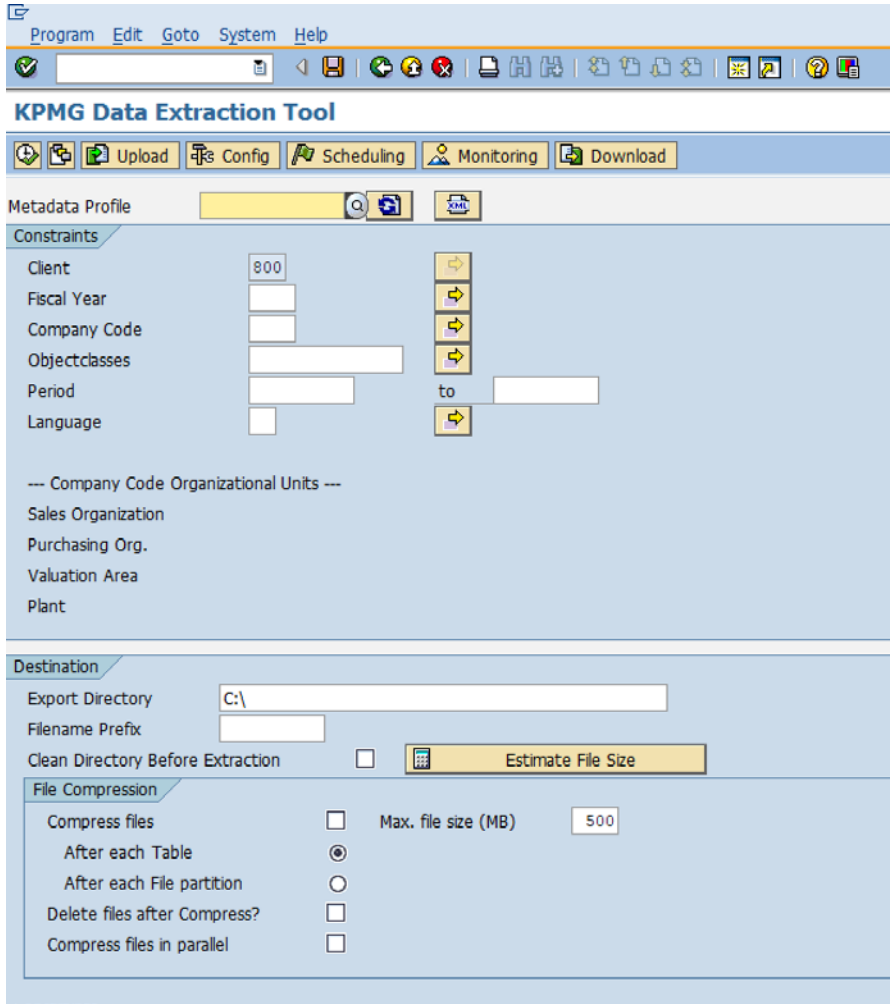
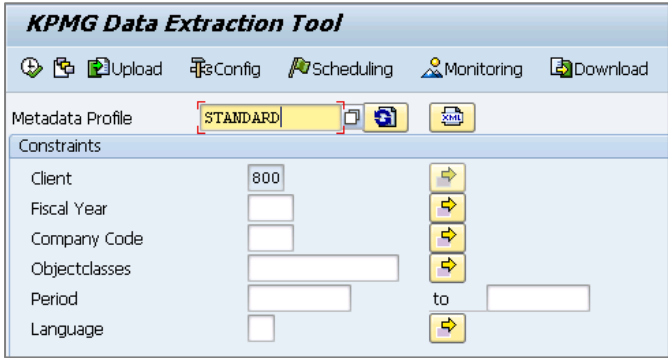
Step	Action
	

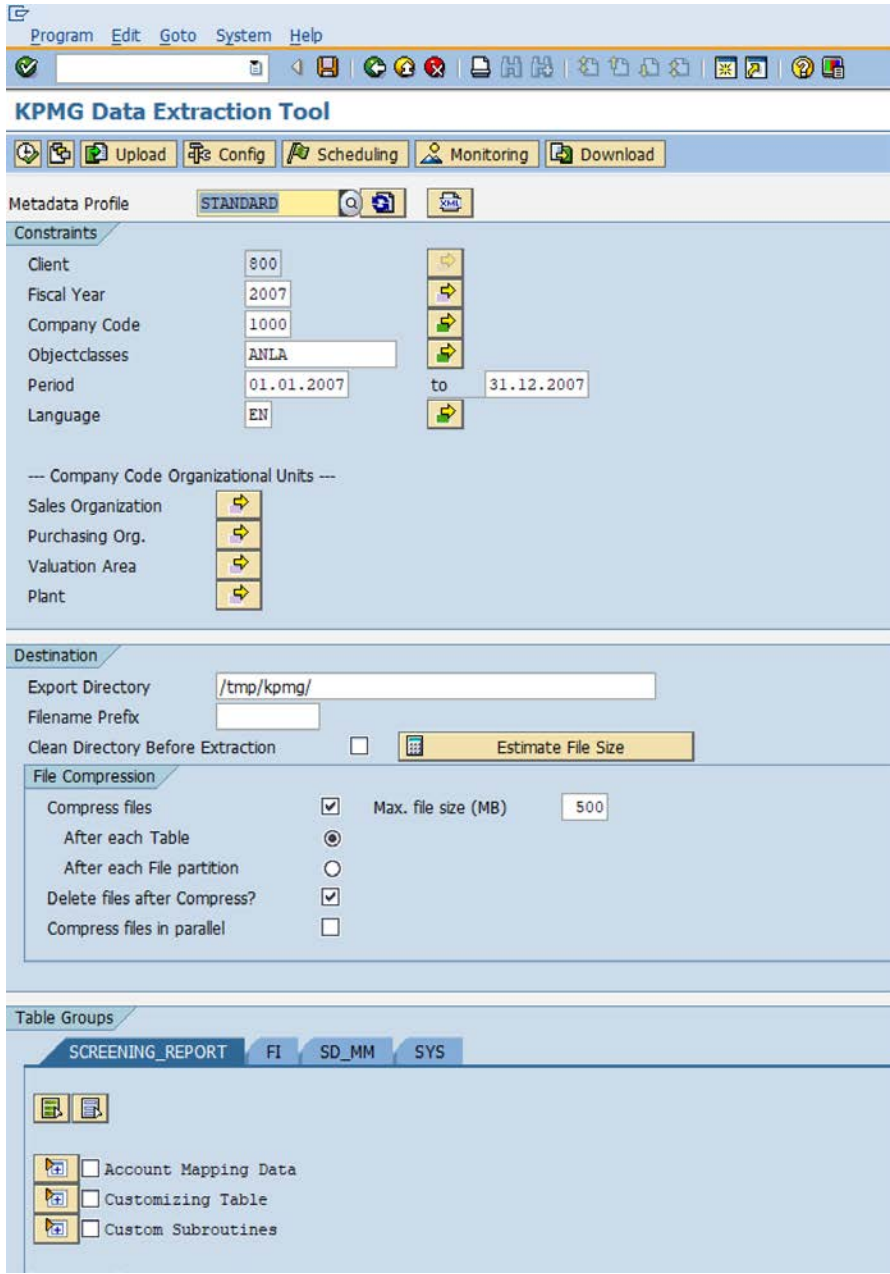
Error Message	Description
– Invalid Company code	– If the Company code used in the XML file is invalid a message is displayed
– “Invalid: Missing/Empty tag”	– Required information may be missing like: 1) <TABNAME> in Table objects; 2) <SUBROUTINEPOOL> or <SUBROUTINE> or <SUBFUNC> in Subroutine objects; 3) <NRFIELD> or <PKGSIZE> in objects using Number range logic; 4) <DAYSPERLOOP> or <PKGSIZE> in objects using Day by Day logic; 5) <PKGSIZE> in objects using Helper table logic;
– “Invalid: Table...in Object”	– Name of table specified in <TABNAME> tag for the Object is not in the SAP Data Dictionary
– “Invalid: Header Table” or “Invalid: Footer Table”	– Name of table specified in <HEADERTABLE> tag is not in the SAP Data Dictionary
– “Invalid: <ConstraintTable...”	– Name of table specified in <CONSTRAINTTABLE> tag is not in the SAP Data Dictionary
– “Invalid: <Field> in <ConstraintTable”	– Name of table field specified under the <CONSTRAINTTABLE> tag is not in the SAP Data Dictionary
– “Improper Bracket Handling In HARDCONSTRAINT”	– Syntax check to make sure that “(“ and “)” come in pairs in <HARDCONSTRAINT> tag
– “Invalid: <Table name=...”	– Name of table specified in <TABLE> tag is not in the SAP Data Dictionary
– “Invalid: <Field> in <Table”	– Name of table field specified under the <TABLE> tag is not in the SAP Data Dictionary

Error Message	Description
– “Invalid: Helper table name”	– Name of table specified in <HELPERTABLE> tag is not in the SAP Data Dictionary

2.2. Configure Screening Data Extraction

The Screening data extraction extracts information regarding the organizational structure, financial account structure, and basic system settings for planning purposes. The output of the Screening data extraction is used to determine the selection parameters and optimal technical settings for the extraction of the transaction data.

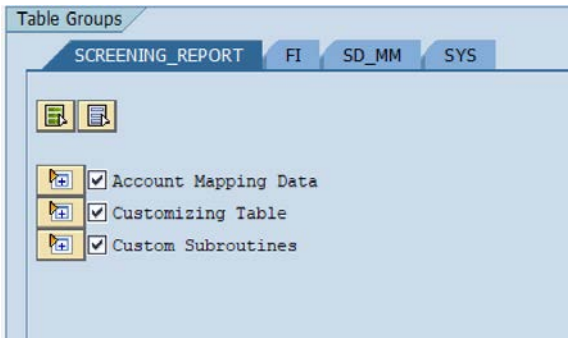
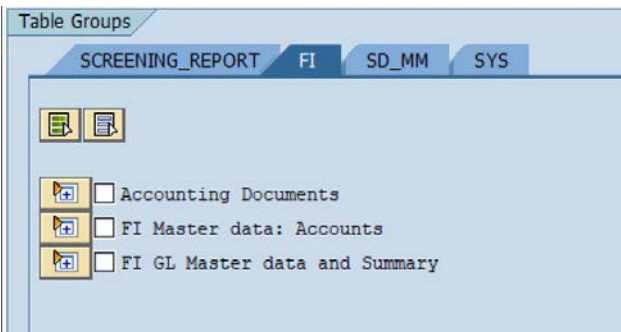
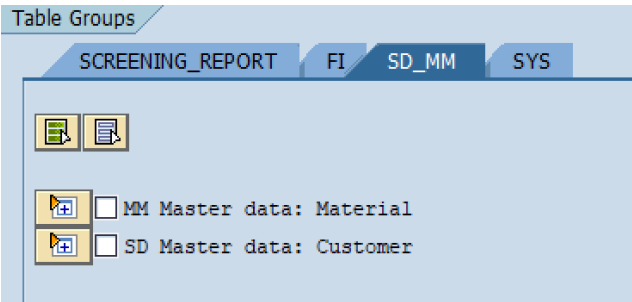
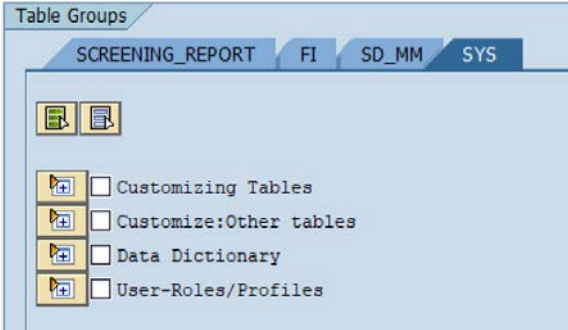
Step	Action
1.	From the landing page, click  From the different other pages, click  Result: Configuration screen displays 
2.	Select Metadata Profile on the top row. 

Step	Action
3.	Click  Result: The values from the Metadata Profile will be populated into the fields. 

The parameters in the selection screen are pre-defined by and loaded in the system as a Metadata Variant. With the not change any fields before discussing your changes with

in the Scope and Configuration file n of the **Export directory** field, do

The four sections in the Data Extraction Tool for the Screening data extraction are described below.

Step	Action
1.	Check all selections on the Table Groups tab 'SCREENING_REPORT' 
2.	Uncheck all selections on the Table Groups tab 'FI' 
3.	Uncheck all selections on the Table Groups tab 'SD_MM' 
4.	Uncheck all selections on the Table Groups tab 'SYS' 

Metadata Profile

The profile is required to populate the Constraints, Table Groups selection and other extraction parameters and specifications.

Constraints

The data selections in this section apply limitations to certain tables in the account mapping data.

Destination

Prior to running the report the output option and Export directory need to be defined. Enter a valid folder or directory path on the application server. Verify that the directory path contains a trailing-slash if it is a Unix environment or a trailing-backslash if it is a Windows environment. Example: For UNIX /usr/sap012/

Ensure that the export directory has access restrictions in place so the extracted data can only be accessible for authorized users.

Destination

Export Directory: /tmp/kpmg/

Filename Prefix:

Clean Directory Before Extraction: ☐

Estimate File Size

File Compression

Compress files: ☒ Max. file size (MB): 500

After each Table: ☒

After each File partition: ☐

Delete files after Compress?: ☒

Compress files in parallel: ☐

Filename Prefix

The Filename Prefix will allow several jobs extracting the same table or tables into the same directory to execute in parallel without overwriting one another.

File Compression

The Compress files checkbox specifies that the extract files need to be compressed. When this option is selected, files will either be compressed after data extraction has completed for the Object (header, footer and related data tables) or right after each file (partition) is closed depending on which of the following 2 options is selected:

- "After each Table"
- "After each File partition"

Selecting the *Delete files after Compress* option just confirms that you want to delete the original extract files to free up space. The parameter *Max. File size (in MB)* is necessary for compression because the whole file to be compressed has to be read into memory and memory is limited which makes it necessary to partition the large files into smaller files. These options are useful for systems with memory constraints or with limited disk space in the application server file directory.

Estimate File Size

If user clicks on the 'Estimate File Size' button, a report is generated with the estimate for number of records for each table and expected file size.

Compress files in Parallel

This option can be used to compress files in parallel to data extraction to improve the extraction time.

Clean Directory Before Extraction

The 'Clean Directory Before Extraction' option provides the possibility to clean the files in a directory from a prior run. Only files are deleted which means sub-directories will be kept. If "Filename Prefix" is specified, only files having this prefix will be deleted.

Table Groups

In the Table Groups selection screen block, there are four report tabs defined. The first tab is the one used for the specifications for the Screening data extraction.

The selected checkboxes and selections define the settings and show the SAP tables and table fields to be downloaded plus the selection options or constraints for the tables. In the special case of a group for extracting data via subroutines instead of table queries, the subroutine names are shown instead of tables. Do not delete checkboxes or change any of the selections without first discussing the changes with

Dynamic date/Period Selection

To use this feature click on the check box 'Activate Dynamic Selection'. Doing so will expand the hidden fields that can be used for this section.

Table Groups

SCREENING_REPORT FI SD_MM SYS

Account Mapping Data ☒

Customizing Table ☒

Custom Subroutines ☒

Dynamic Date/Period Selection

Activate Dynamic Selection ☐

Technical Settings

Show technical configuration options

This feature can be used to schedule extraction jobs on a regular basis (e.g. daily, monthly, yearly) without changing the variant. This allows to update the date constraints for the selection based on the current system date or current period start date. If users want to change the extraction period date dynamically they can either select the radio button Date-Based Selection or Period-Based Selection. Negative values will shift the date backwards and positive values will shift it forwards.

Dynamic Date/Period Selection

Activate Dynamic Selection ☒

☐ Date-Based Selection(Start/End)

Years /

Months /

Days /

☒ Period-Based Selection(Start/End)

Years /

Months /

Technical Settings

Additional extraction options are available primarily provided to achieve better performance and also to give flexibility in order to avoid technical problems such as memory shortage.

Technical Settings

☐ Hide technical configuration options

Extraction Logic | Package size | Number Range Min/Max | Archive Settings

FI
SD_MM
SYS

Partition file size limit (MB)

Column separator

of PID for Parallel processing

☐ Output Job log for Parallel processing

☒ Remove Linefeed characters in Text fields

☒ Download related purchase/sales transactions ☒ Duplicate Removal(Memory Intense)

☐ Download recursively all sales documents related to CY documents

☐ Include all recursive data of sched. agreements with indiv calls

☐ Download related purchase documents for PTP

☒ Download related change documents

☒ Download related accounting documents

☒ Query Footer tables outside Header SQL for Simple query logic

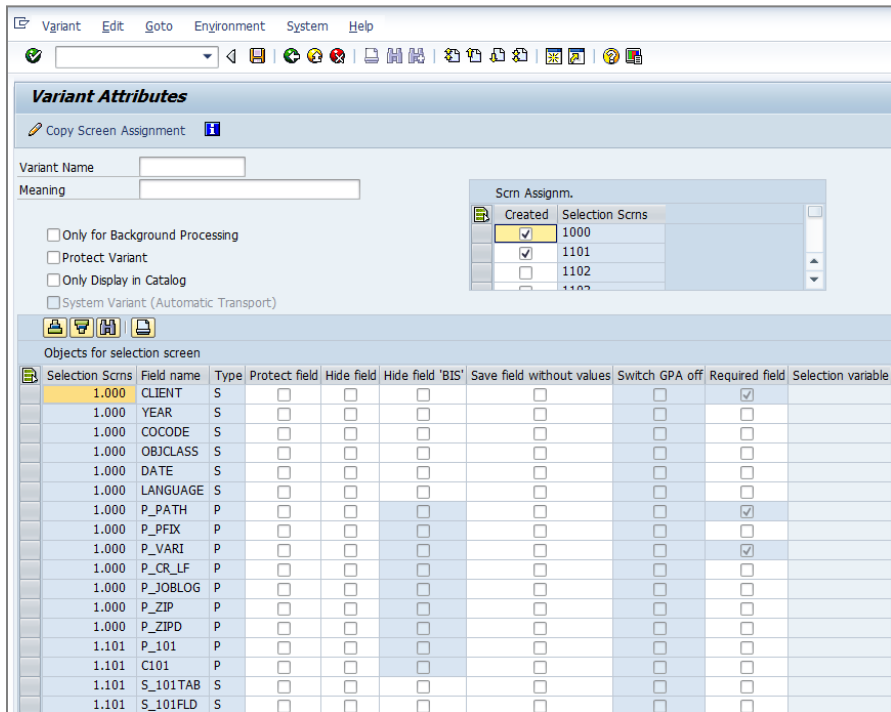
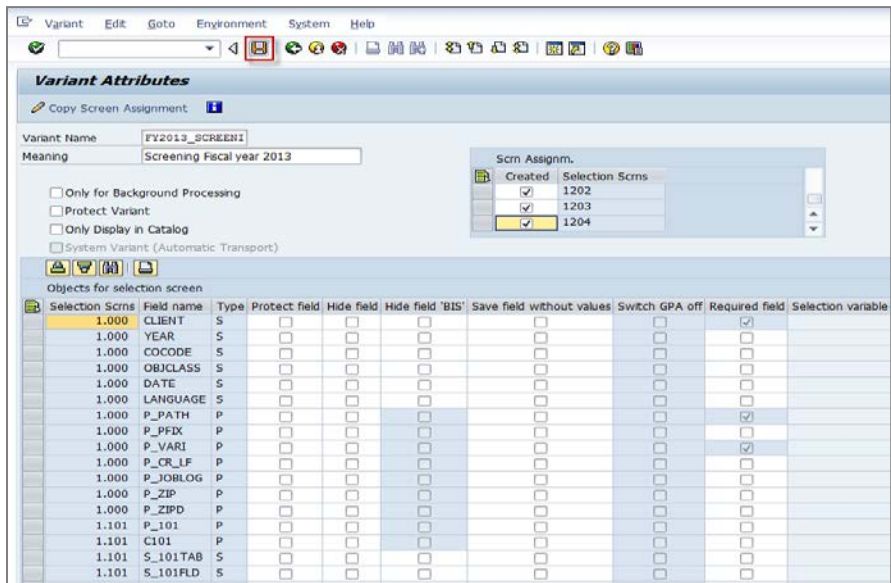
☐ Include archived Master/Helper table data

☐ Delete all data in MD5 Hash DB table

Remove Linefeed characters in Text field option

This option is selected by default to remove all Linefeed characters that may be the content of a data field as they lead to a line break in the exported text file which causes errors during the import process by the missing end of record delimiter. Usually these characters are found in description fields. There is the option to deselect this feature to increase the extraction runtime and reduce the load on the SAP system. All the other Technical Settings are not relevant for the Screening data extraction.

An ABAP program variant (or Selection screen Variant) is required to create a Background job to execute the Data Extraction Tool. It is important to note that because the Data Extraction Tool includes several sub screens, unless all these sub screens are displayed on screen before saving the selection screen to create the variant, some of the sub screens may not be saved/ included with the variant. It is very important to make sure that all the screens are selected/ checked before saving.

Step	Action
	From the configuration screen, click Result: Variant Attributes displays 
2.	Enter desired Variant ID in the Variant Name field and Description in the Meaning field. Make sure all the checkboxes in the Screen Assignment are selected/ checked. Use the scroll bar to see all the screens in the list. 

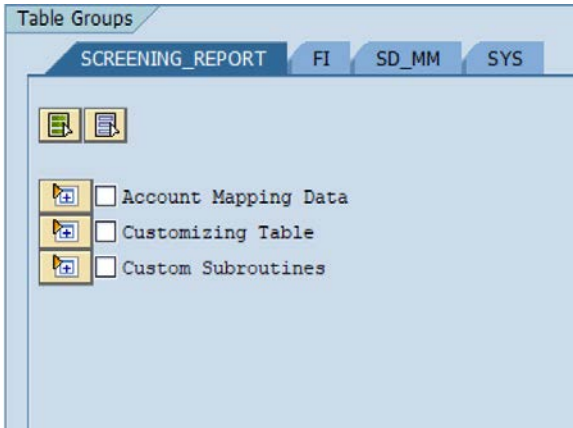
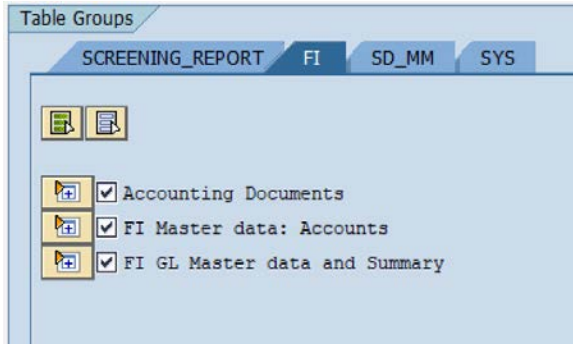
Step	Action
	Click Result: Message confirming that the Variant is saved displays

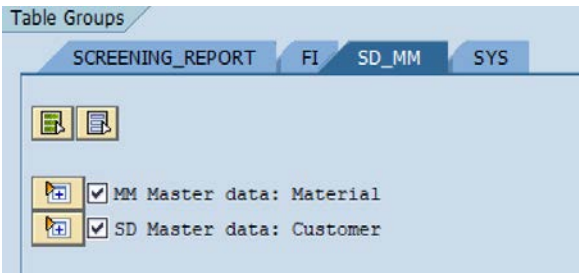
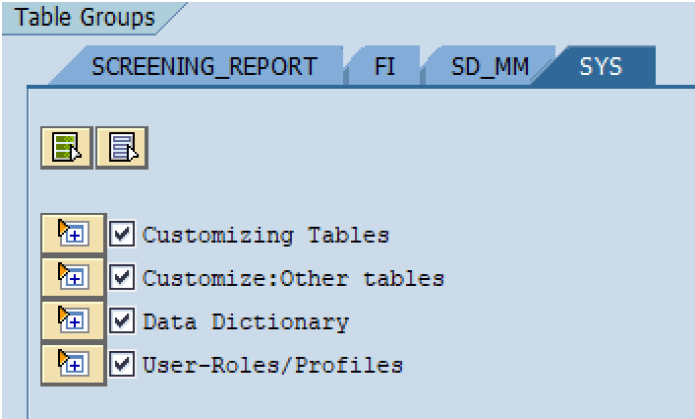
2.3. Configure Transaction Data Extraction

There are three report tabs used to extract transaction and master data from the SAP system. Due to the high volume of the tables downloaded by these reports it is generally necessary to schedule them separately as background jobs.

- FI tab extracts data from the Financial module
- SD_MM tab extracts data from the Sales and Distribution and Materials Management modules
- SYS tab extracts system data and change documents

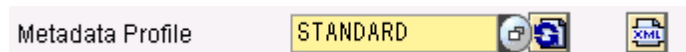
The parameters in the selection screen are pre-defined by **directory** field, do not change any fields before discussing With the e of the **Export** anges with

Step	Action
1.	Uncheck all selections on the Table Groups tab 'SCREENING_REPORT'. 
2.	Check all selections on the Table Groups tab 'FI'. 

Step	Action
	Check all selections on the Table Groups tab 'SD_MM'. 
4.	Check all selections on the Table Groups tab 'SYS'. 

Metadata Profile

The profile is required to populate the Constraints, Table Groups selection and other extraction parameters and specifications.



The three main sections in the ABAP Data Extraction Tool for the transaction data extraction are described below:

Constraints

The Constraints section defines the limitations that are applied when downloading the transaction data tables.

The **Objectclasses** field is used only to restrict the extraction of the change document tables.

The **Period** fields' dates are pre-initialized periods which usually exceed the actual period under consideration by one month in each direction to cover business processes beginning earlier and potentially ending through the day of extraction. This also applies to the fiscal year parameters which contain the subsequent financial year transactions.

The **Sales organization**, **Purchasing organization**, **Valuation area** and **Plant** constraints are determined by the Company code constraints based on the current Enterprise Structure assignments. These four organizational unit constraints are determined internally when the report is executed so it's usually not necessary to specify the values in the selection screen. However, there may be instances where the current assignments are not valid for the extraction/reporting period. Some units may have been re-assigned to other company codes, or new units added, etc. If this is the case, these constraints have to be maintained to reflect the organization during the reporting period before executing the report.

Destination

Prior to running the report the output option and Export directory need to be defined. Enter a valid folder or directory path on the application server. Verify that the directory path contains a trailing-slash if it is a Unix environment or a trailing-backslash if it is a Windows environment. Example: For UNIX /usr/sap012/

Ensure that the export directory has access restrictions in place so the extracted data can only be accessible for authorized users.

The screenshot shows a software configuration window. The 'Destination' section includes an 'Export Directory' field with the value '.', a 'Filename Prefix' field, and a 'Clean Directory Before Extraction' checkbox. A yellow button labeled 'Estimate File Size' is also present. The 'File Compression' section contains several options: 'Compress files' (checked), 'After each Table' (selected with a radio button), 'After each File partition' (radio button), 'Delete files after Compress?' (checked), and 'Compress files in parallel' (checkbox). A 'Max. file size (MB)' field is set to '500'.

Filename Prefix

The Filename Prefix will allow several jobs extracting the same table or tables into the same directory to execute in parallel without overwriting one another.

Estimate File Size

This helps the User to know the estimate size of data files and the Number of records of data files which will be extracted.

Compress files in parallel

Compression of files can happen in parallel to data extraction when this option is selected.

Compress files and Delete files after Compress options

These options are useful for systems with limited disk space in the application server file directory. The option exist to compress each file after it got created or the files for each table after the table got extracted. Selecting the *Delete files after Compress* option just confirms that you want to delete the original extract files to free up space. The parameter *Max. File size (in MB)* is necessary for compression because the whole file to be compressed has to be read into memory and memory is limited which makes it necessary to partition the large files into smaller files.

Clean Directory Before Extraction

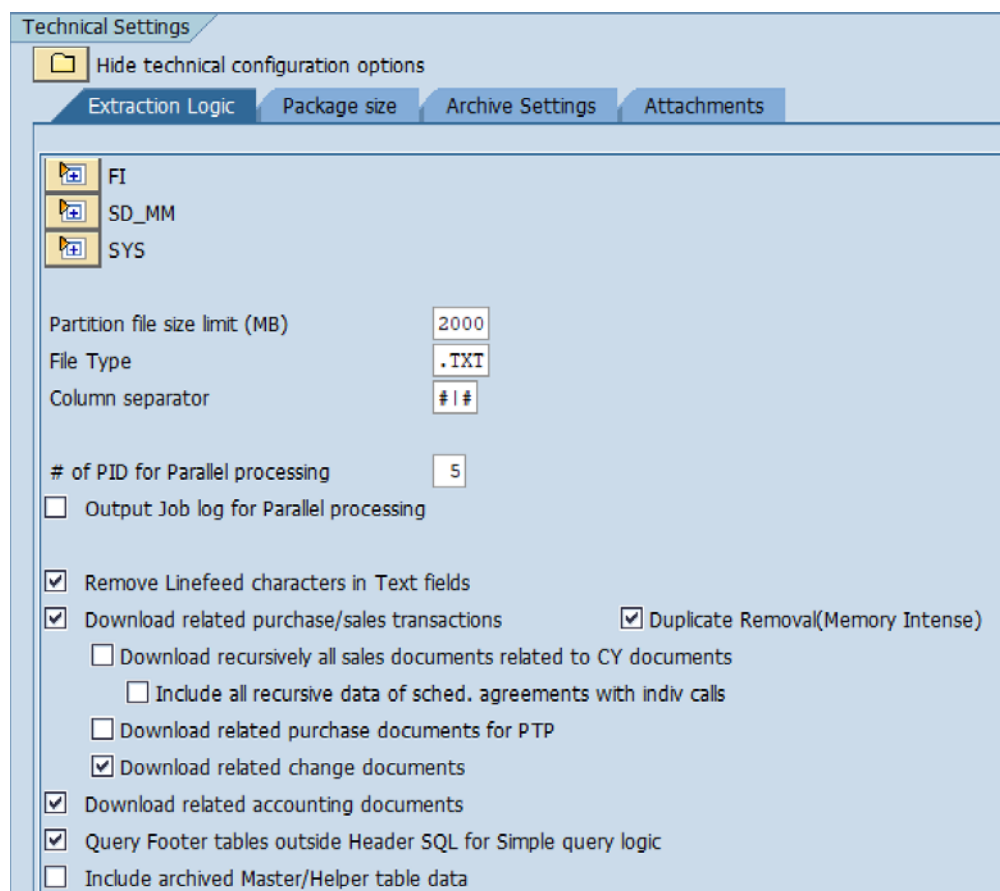
The 'Clean Directory Before Extraction' option provides the possibility to clean the files in a directory from a prior run. Only files are deleted which means sub-directories will be kept. If "Filename Prefix" is specified, only files having this prefix will be deleted.

Table Groups

In the Table Groups selection screen block, there are four report tabs defined. The first tab is the one used for the specifications for the Screening data extraction. The second tab is for extracting FI data, the third for SD_MM data, and the fourth for System data. The selected checkboxes and selections define the settings and show the SAP tables and table fields to be downloaded plus the selection options or constraints for the tables. In the special case of a group for extracting data via subroutines instead of table queries, the subroutine names are shown instead of tables. Do not d the checkboxes or change any of the selections without first discussing the changes with

Technical Settings

Additional extraction options are available primarily provided to achieve better performance and also to give flexibility in order to avoid technical problems such as memory shortage.



Partition file size limit (MB)

This field sets the maximum size of the extracted files so that if the data to be extracted for a table exceed this size, they will have to be split into partition files with each file no larger than this file size limit. 2GB is the default but is changeable.

Column separator

The Column separator is a 3-character field that will be inserted between two columns (or fields) in the extract file.

File Type

The field gives an option to select the extracted file extension i.e. .TXT or .CSV. If .CSV file type is selected, the field separator is updated to be comma (,).

of PID for Parallel processing option

This parameter is relevant only when using the Parallel processing option and sets the limit of the number of parallel DIA processes that can be active at any one time to prevent the extraction job from occupying all the available DIA processes.

Output Job log for Parallel processing option

This option may only be useful when it's necessary to track the events and timing when using the Parallel processing option. Otherwise, leave it deselected as it will create a lot of job log messages that may create spool issues.

Remove Linefeed characters in Text fields option

Inter
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Cooperative (International
arties, nor does

International"), a Swiss entity. Member firms of the
o client services. No member firm has any autho
International have any such authority to obligate or bind any member f

network of in
ligate or bind
any member f

nt firms are affiliated with
International or any other
rights reserved.

This option is selected by default to remove all Linefeed characters that may be part of the content of a data field as they can be confused for an end of record delimiter. Usually these are found in some long text fields. There is no good reason to deselect this option.

Download related transactions option

In addition, there is a special provision to extract related transactions and the related tables will also be added to the list of tables to extract. These related tables are always displayed on the selection screen (hardcoded, meaning not via XML configuration) but will only be extracted if the corresponding checkboxes are selected. This option is selected by default for the purpose of extracting older data or documents related to the Purchase invoices, Sales invoices, Accounting documents and Change documents. For example if the Sales order related to the Sales invoice is older than the current reporting period, it won't otherwise be available in the KAAP application for analysis because the Sales document extraction will only extract Sales orders for the current reporting period. In addition to the above dependent transaction logic for sales subledger documents outside of the analysis period has been expanded with the following:

- An option to extract recursively all documents preceding sales order documents (e.g. contracts, scheduling agreements)
- An option to extract all the transaction documents (e.g. sales orders, deliveries, invoices, etc.) with individual calls related to those newly extracted contracts or scheduling agreements
- An option to extract invoice related purchase documents for PTP

Query Footer tables outside Header SQL for Simple query logic

This option when selected will extract footer table data only after the header data extraction is completed instead of within the header table looping to prevent (DB2) database timeouts due to the header table SQL taking longer than the set maximum. The footer extraction use the document number range as determined in the head table data selection. Likely drawback is that some footer table data may be extracted that do not have corresponding extracted header table data. <DOCNBR> in the XML file is used to remove the hardcoded settings by flexible XML tags that can be altered depending on the field to query by.

Delete all data in MD5 Hash DB table

This option if selected refreshes the table A/MD5_20. This table is used in 2020 extraction tool to add the feature of data anonymizati

Technical Configuration options:

Extraction Logic

The screenshot shows the 'Technical Settings' window with the 'Extraction Logic' tab selected. The window has a 'Hide technical configuration options' button at the top left. Below it are four tabs: 'Extraction Logic', 'Package size', 'Archive Settings', and 'Attachments'. The 'Extraction Logic' tab contains a tree view on the left with folders 'FI', 'SD_MM', and 'SYS'. Under 'FI' is 'Accounting Documents' (containing 'BKPF') with a checkbox for 'Activate Parallel processing' and a dropdown for 'DaybyDay Date-Delimited' set to '3'. Under 'SYS' is 'Customize:Other tables' (containing 'SWW_WI2OBJ') with a dropdown for 'Helper table' set to '4'. Below the tree are several configuration fields: 'Partition file size limit (MB)' set to '2000', 'File Type' set to '.TXT', 'Column separator' set to '#|#', '# of PID for Parallel processing' set to '5', and a checkbox for 'Output Job log for Parallel processing' which is unchecked. At the bottom are several checkboxes: 'Remove Linefeed characters in Text fields' (checked), 'Download related purchase/sales transactions' (checked), 'Duplicate Removal(Memory Intense)' (checked), 'Download recursively all sales documents related to CY documents' (unchecked), 'Include all recursive data of sched. agreements with indiv calls' (unchecked), and 'Download related purchase documents for PTP' (unchecked).

Certain tables are identified to better be handled using a specific Extraction logic other than a simple/flat query. If for some reason, a simple query is desired instead, there is an option provided to disable using the special extraction logic by removing the extraction logic code for the table.

Parallel processing will improve performance for the larger tables where the overhead required to perform parallel processing is outweighed by the amount of time required to extract all the data sequentially. Parallel processing may be activated for these tables by marking their corresponding checkbox.

Package size

Technical Settings

Hide technical configuration options

Extraction Logic Package size Archive Settings Attachments

Adjust pkg size

FI

Accounting Documents

BKPF 3637

Subtables 364

FI Master data: Accounts

SKA1 50000 SKAT 50000

FI GL Master data and Sum

FAGLFLEXT 32715 FAGL_ACTI 50000

SD_MM

SYS

Due to the size of the tables, an iterative approach is taken whereby each group of tables is downloaded one at a time. The logic that is used to download tables can be customized with **Package sizes** (number of records to be extracted in one iteration) to comply with available memory as well as parallel processing settings to execute multiple parallel download processes. Depending upon the basic system settings and available memory, Upload program can change the package size values prior to the extraction run.

Archive Settings

Technical Settings

Hide technical configuration options

Extraction Logic Package size Number Range Min/Max Archive Settings

Tables for export	Arch. Index	Pkg Size
☐ Accounting Documents	ZARIXFI2	100
☐ Purchase Requisition Doc	ZARIXMM2	100
☐ Purchase Documents	ZARIXMM3	100
☐ Material Movement Doc	ZARIXMM4	100
☐ Invoice Documents	ZARIXMM5	100
☐ Delivery Documents	ZARIXSD5	100
☐ Sales Documents	ZARIXSD3	100
☐ Sales Invoice Documents	ZARIXSD4	100

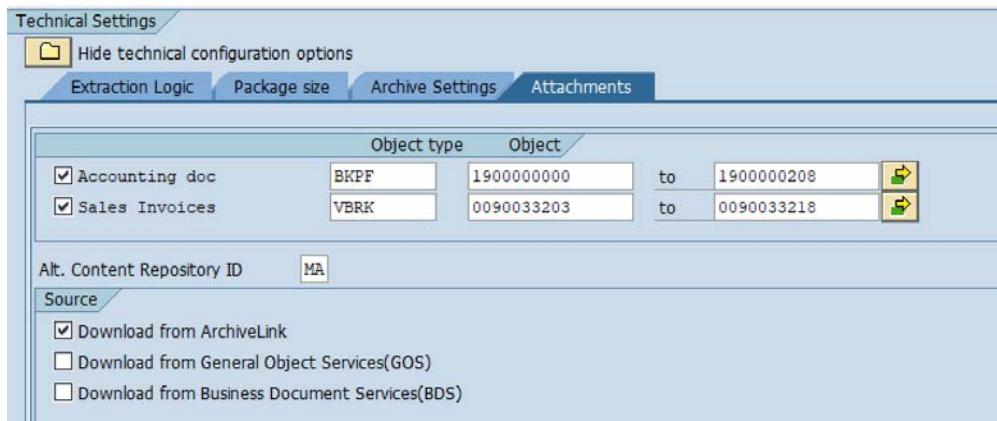
Those settings can be checked to extract data from the SAP archive. The Table Groups that are then related to several Archiving Index can be checked and the package size can get changed.

Attachments

Used to download the documents attached to business objects in SAP.

The Config program reads the Metadata profile, updates the Attachments tab on screen with values, reads user changes to this screen, validates user input and saves the profile information.

Multiple SAP object types are supported and user can enter multiple number ranges for each Object. Up to a maximum of 8 Object types can be entered in the XML file. An attachment log file is generated for each Object type that has the list of documents extracted.



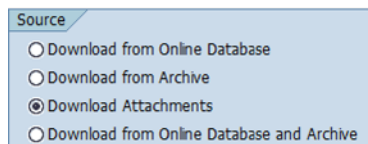
Object type	Object
☒ Accounting doc	BKPF 1900000000 to 1900000208
☒ Sales Invoices	VBRK 0090033203 to 0090033218

Alt. Content Repository ID: MA

Source:

- ☒ Download from ArchiveLink
- ☐ Download from General Object Services(GOS)
- ☐ Download from Business Document Services(BDS)

Source



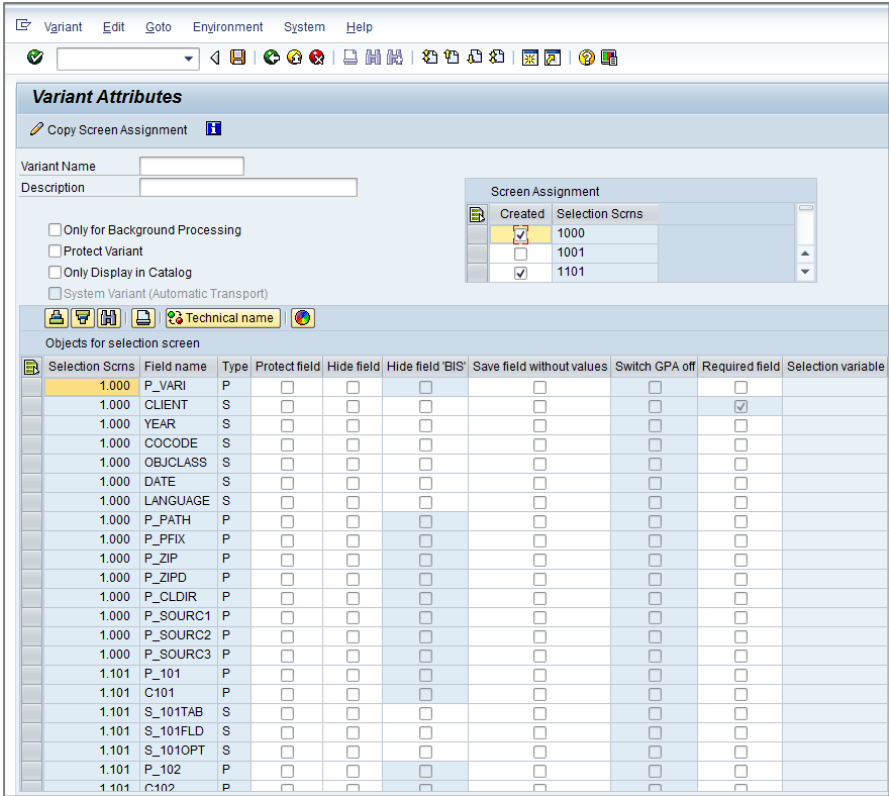
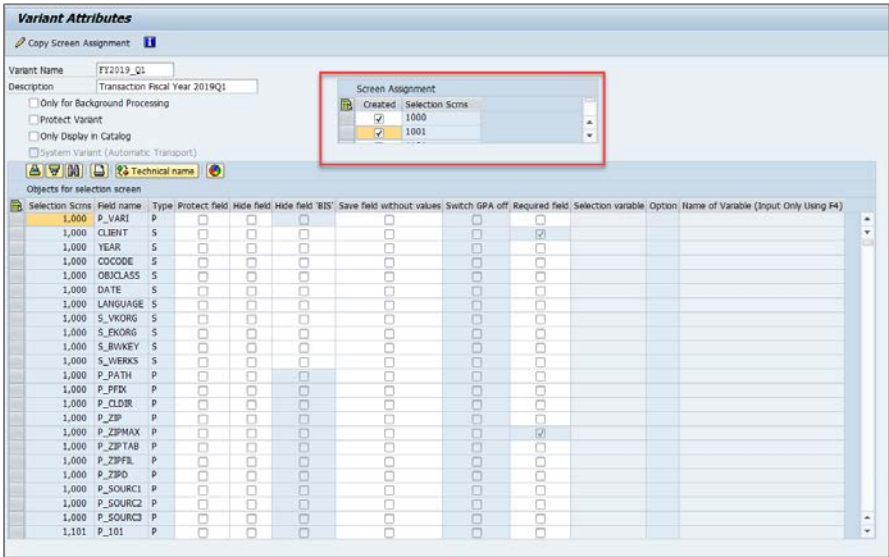
Source

- ☐ Download from Online Database
- ☐ Download from Archive
- ☒ Download Attachments
- ☐ Download from Online Database and Archive

This option is to choose between extracting data from the Online Database and/or the Archive or Attachments. In case of the Archive data extraction the checkboxes for the Table Groups have to be checked on the Archive Settings.

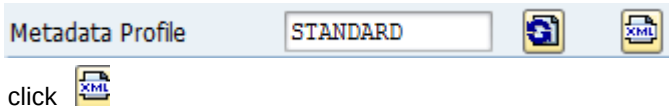
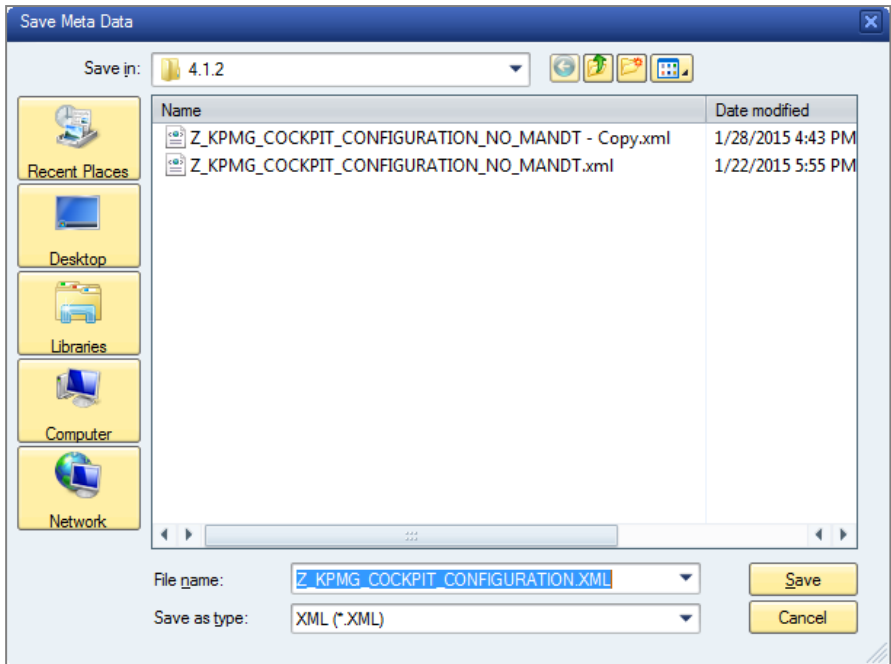
For details regarding the available parameter settings, refer to the Appendix in this document.

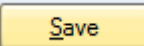
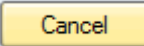
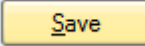
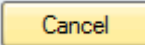
An ABAP program variant (or Selection screen Variant) is required to create a Background job to execute the Data Extraction Tool. It is important to note that because the Data Extraction Tool includes several sub screens, unless all these sub screens are displayed on screen before saving the selection screen to create the variant, some of the sub screens may not be saved/ included with the variant. It is very important to make sure that all the screens are selected/ checked before saving.

Step	Action
	From the configuration screen, click  Result: Variant Attributes displays. 
2.	Enter desired Variant ID in the Variant Name field and Description in the Meaning field. Import: Make sure all the checkboxes in the Screen Assignment are selected/ checked. Use the scroll bar to see all the screens in the list. 

Step	Action
	Click Result: Message confirming that the Variant is saved displays

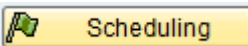
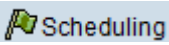
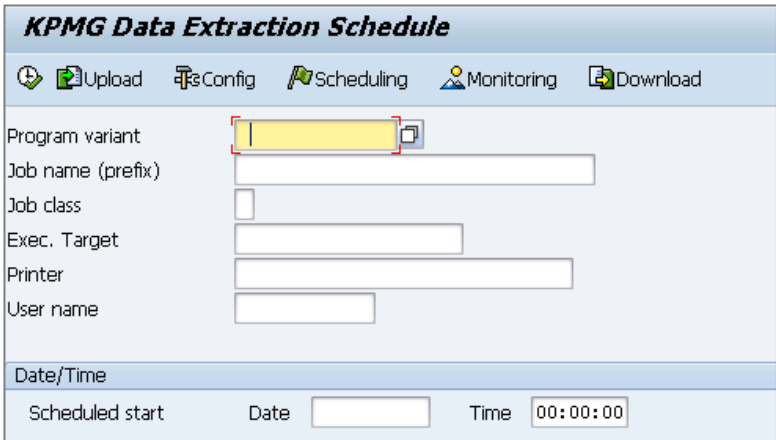
The Metadata Profile, after initially importing into the Data Extraction Tool selection screen, can be modified and re-configured in the selection screen: package sizes may be optimized, extraction logic may be switched to the option that works best for the client, etc. There is an option to save this new configuration by generating a new Scope and Configuration file.

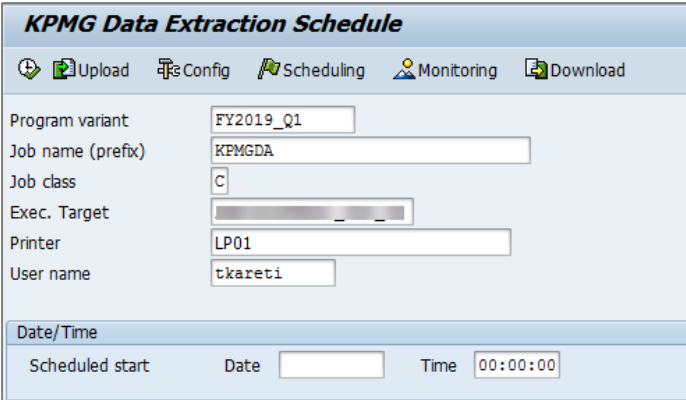
Step	Action
1.	From the selection screen,  Result: Pop-up window displays to prompt user to specify file name and location for the generated XML file. 

Step	Action
	Select  or . Result:  saves the generated XML file into the specified file name and location.  aborts the process.

Scheduling Data Extraction

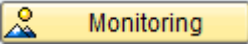
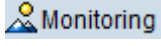
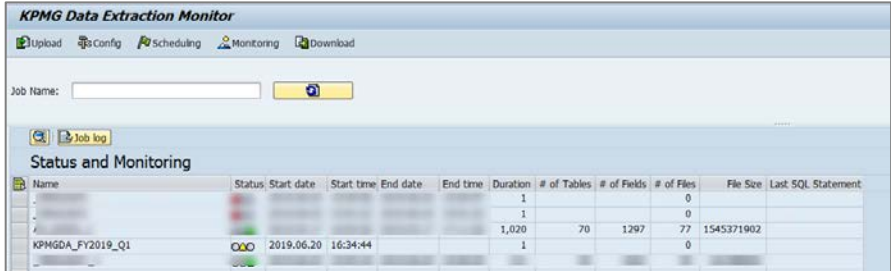
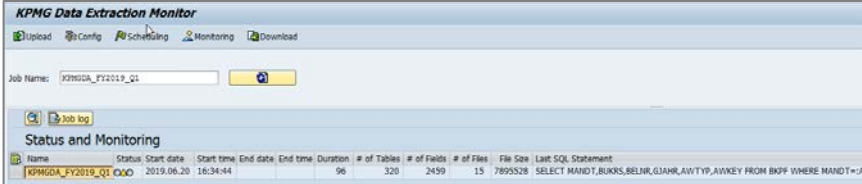
The scheduling program helps with scheduling the extraction job in the background after a variant of the Data Extraction Tool was successfully created in the Configuration step. The following steps have to get executed to schedule the data extraction.

Step	Action
1.	From the landing page, click . From the different other pages, click . Result: Scheduling screen displays 
2.	Select the variant named we stored in the Configuration step. All stored variants can be selected via F4 functionality.

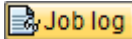
Step	Action
	The following fields should be entered: The Program variant should contain a valid Data Extraction Tool variant, for example, e.g. FY2020_Q1. The Job name (prefix) should contain a prefix for naming the job. The name of the job to be scheduled will be a concatenate the Job name (prefix) e Program variant, e.g., with prefix A, results in job name A_FY2020_Q1. The following remaining fields should only be entered if it is necessary to override the system defaults: The Job class determines the priority of the background job a. High priority b. Middle priority c. Low priority (default) The Exec. Target provides the possibility to specify the processing server used to execute the background job for SAP systems with more than one application servers. The Printer selections provide the possibility to enter a printer ID for the output of the job. The program also allows the user to enter the User name under which the job should be executed. Make sure that the User name has the privileges to execute the job. The remaining selection is the Date/Time when the job should get executed. If the date and time is not specified, the job will be scheduled immediately. 
4.	Click  Result: The extraction job will be scheduled in background.

2.4. Monitoring Data Extraction

The monitoring program allows the user to monitor the extraction job. During the run of the extraction job the user can check manually for the progress of the extraction by entering the monitoring program. The program consists out of an overview screen and a detailed screen, which then provides further details. The following steps should be followed to monitor the extraction jobs.

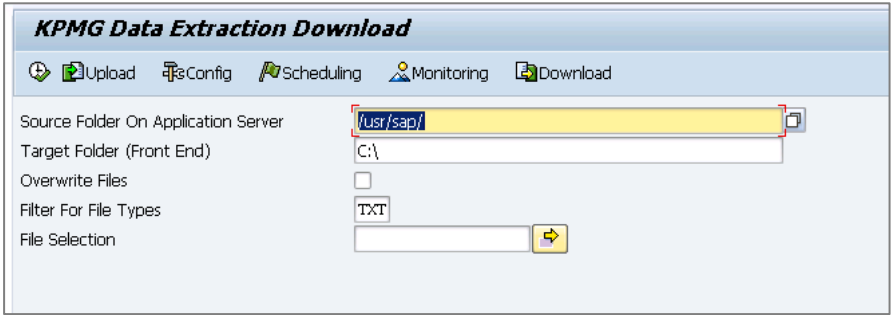
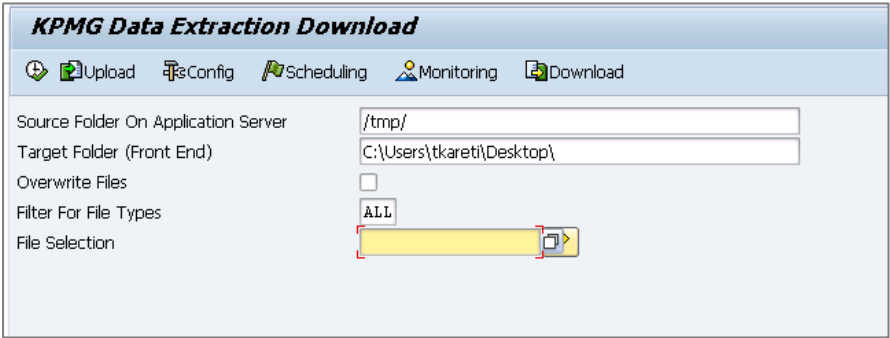
Step	Action
1.	From the landing page, click  From the different other pages, click  Result: Monitoring program screen displays 
2.	Select the Job Name we stored in the Scheduling step. The available background jobs can be selected via F4 functionality. The screen shows the status of the extraction job, start date and time, end date and time, the amount of tables and fields we are going to extract and the last SQL statement. 
3.	Click  Result: The status of the extraction job will be refreshed and loaded.

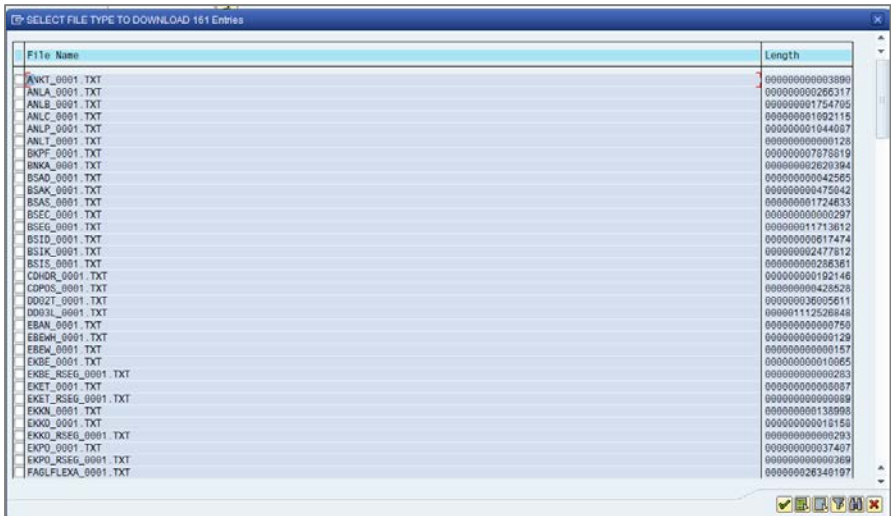
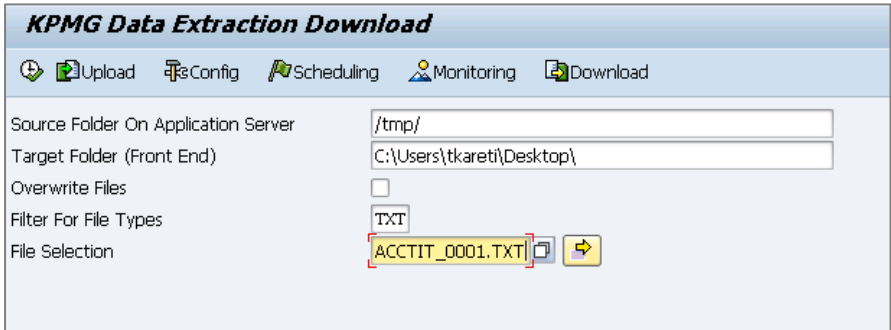
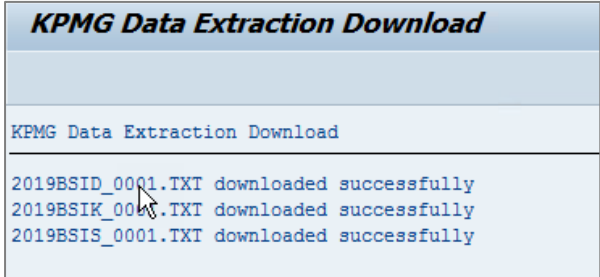
Step	Action																																																																																																
	To reach the extraction detail screen mark the line with the Job Name and click . Result: The job details will be shown that list all the extracted tables and files which were created including information about the size, rows and extraction logic. KPMG Data Extraction Monitor Upload Config Scheduling Monitoring Download Job Name: KPMGDA_FY2019_Q1 Job Details <table> <tr> <th>Name</th><th>Table</th><th>File Name</th><th>Size</th><th>Execution Logic</th><th>Rows</th></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>FAGLFLEX</td><td>2019CD_SAP_FAGLFLEX_0001.TXT</td><td>2501222</td><td>Simple Query</td><td>14,186</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>FAGL_ACTIVEC</td><td>2019CD_SAP_FAGL_ACTIVEC_0001.TXT</td><td>55</td><td>Simple Query</td><td>1</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>GLTO</td><td>2019CD_SAP_GLTO_0001.TXT</td><td>1205861</td><td>Simple Query</td><td>6,683</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>SKA1</td><td>2019CD_SAP_SKA1_0001.TXT</td><td>572084</td><td>Simple Query</td><td>12,431</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>SKAT</td><td>2019CD_SAP_SKAT_0001.TXT</td><td>2471746</td><td>Simple Query</td><td>39,359</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>SKB1</td><td>2019CD_SAP_SKB1_0001.TXT</td><td>81270</td><td>Simple Query</td><td>2,247</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>T011</td><td>2019CD_SAP_T011_0001.TXT</td><td>5677</td><td>Simple Query</td><td>68</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>T001</td><td>2019CD_SAP_T001_0001.TXT</td><td>13801</td><td>Simple Query</td><td>143</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>T009</td><td>2019CD_SAP_T009_0001.TXT</td><td>636</td><td>Simple Query</td><td>23</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>T009B</td><td>2019CD_SAP_T009B_0001.TXT</td><td>72273</td><td>Simple Query</td><td>2,113</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>I011Z</td><td>2019CD_SAP_I011Z_0001.TXT</td><td>278444</td><td>Subroutine</td><td>4,638</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>RSPARAM</td><td>2019CD_SAP_RSPARAM_0001.TXT</td><td>136283</td><td>Subroutine</td><td>1,840</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>X011P</td><td>2019CD_SAP_X011P_0001.TXT</td><td>86160</td><td>Subroutine</td><td>1,735</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>X011Q</td><td>2019CD_SAP_X011Q_0001.TXT</td><td>412326</td><td>Subroutine</td><td>7,387</td></tr> <tr> <td>KPMGDA_FY2019_Q1</td><td>X011S</td><td>2019CD_SAP_X011S_0001.TXT</td><td>57690</td><td>Subroutine</td><td>1,646</td></tr> </table>	Name	Table	File Name	Size	Execution Logic	Rows	KPMGDA_FY2019_Q1	FAGLFLEX	2019CD_SAP_FAGLFLEX_0001.TXT	2501222	Simple Query	14,186	KPMGDA_FY2019_Q1	FAGL_ACTIVEC	2019CD_SAP_FAGL_ACTIVEC_0001.TXT	55	Simple Query	1	KPMGDA_FY2019_Q1	GLTO	2019CD_SAP_GLTO_0001.TXT	1205861	Simple Query	6,683	KPMGDA_FY2019_Q1	SKA1	2019CD_SAP_SKA1_0001.TXT	572084	Simple Query	12,431	KPMGDA_FY2019_Q1	SKAT	2019CD_SAP_SKAT_0001.TXT	2471746	Simple Query	39,359	KPMGDA_FY2019_Q1	SKB1	2019CD_SAP_SKB1_0001.TXT	81270	Simple Query	2,247	KPMGDA_FY2019_Q1	T011	2019CD_SAP_T011_0001.TXT	5677	Simple Query	68	KPMGDA_FY2019_Q1	T001	2019CD_SAP_T001_0001.TXT	13801	Simple Query	143	KPMGDA_FY2019_Q1	T009	2019CD_SAP_T009_0001.TXT	636	Simple Query	23	KPMGDA_FY2019_Q1	T009B	2019CD_SAP_T009B_0001.TXT	72273	Simple Query	2,113	KPMGDA_FY2019_Q1	I011Z	2019CD_SAP_I011Z_0001.TXT	278444	Subroutine	4,638	KPMGDA_FY2019_Q1	RSPARAM	2019CD_SAP_RSPARAM_0001.TXT	136283	Subroutine	1,840	KPMGDA_FY2019_Q1	X011P	2019CD_SAP_X011P_0001.TXT	86160	Subroutine	1,735	KPMGDA_FY2019_Q1	X011Q	2019CD_SAP_X011Q_0001.TXT	412326	Subroutine	7,387	KPMGDA_FY2019_Q1	X011S	2019CD_SAP_X011S_0001.TXT	57690	Subroutine	1,646
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KPMGDA_FY2019_Q1	GLTO	2019CD_SAP_GLTO_0001.TXT	1205861	Simple Query	6,683																																																																																												
KPMGDA_FY2019_Q1	SKA1	2019CD_SAP_SKA1_0001.TXT	572084	Simple Query	12,431																																																																																												
KPMGDA_FY2019_Q1	SKAT	2019CD_SAP_SKAT_0001.TXT	2471746	Simple Query	39,359																																																																																												
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KPMGDA_FY2019_Q1	T011	2019CD_SAP_T011_0001.TXT	5677	Simple Query	68																																																																																												
KPMGDA_FY2019_Q1	T001	2019CD_SAP_T001_0001.TXT	13801	Simple Query	143																																																																																												
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KPMGDA_FY2019_Q1	T009B	2019CD_SAP_T009B_0001.TXT	72273	Simple Query	2,113																																																																																												
KPMGDA_FY2019_Q1	I011Z	2019CD_SAP_I011Z_0001.TXT	278444	Subroutine	4,638																																																																																												
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KPMGDA_FY2019_Q1	X011Q	2019CD_SAP_X011Q_0001.TXT	412326	Subroutine	7,387																																																																																												
KPMGDA_FY2019_Q1	X011S	2019CD_SAP_X011S_0001.TXT	57690	Subroutine	1,646																																																																																												

5.	To reach the job log of the extraction job mark the line with the Job Name and click . Result: The job log will be shown that list all the steps of the extraction job. 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2.5. Download Data Extracts

The download program provides the possibility to load the extracted data from the target directory of the application server to the clients' machine without further interaction with the basis team. Several files can get selected and copied than through the SAP application layer to the client.

Step	Action
1.	From the landing page, click  Download From the different other pages, click  Download Result: Download program screen displays 
2.	Select the source folder on application server by using the F4 functionality or directly entering the destination path of the extraction job. In addition select the Target Folder on the clients' machine.  The possibility to overwrite files or filter for certain file types is possible. Hit ENTER key to confirm change in the folder names.

Step	Action
	To select the files to be copied, click into the File Selection field and then press F4 key and the list of files found in the 'Source Folder On Application Folder'.  Select/check the files to be downloaded. Please refrain from selecting too many files as downloading files is an online execution of the program and hence restricted to the Timeout parameter of the system. To change the selected files please click  and then select or unselect files
4.	To initiate the download click   The download will be executed as a front-end transaction. Please be aware that during that time no further transaction is possible on that session. List of files downloaded are displayed in the report output. 

3. Troubleshooting

3.1. Possible errors

If the extraction program is interrupted or cancelled, send the following information via email to the team.

1. Details of the ABAP-Runtime error of the cancelled extraction report as shown in SAP transaction ST22.
2. Runtime of the extraction job (in seconds) up to the point where the report failed as shown in SAP transaction SM37.
3. Name and data volume of the files that have been created in the extraction folder up to the point where the report failed.
4. Action log files that have been created in the extraction folder (files starting with LOG_*).

If the team asks you to provide any test results after a successful execution of the reports in the comments, please provide the items listed in points 2 to 4 above.

What could go wrong?

Execution of the Reports

ISSUE: The execution of an ABAP report fails because the report fails to write any data in the extraction folder.

Solution 1:	Verify that the SAP user executing the ABAP reports has sufficient write permissions for the extraction folder on the file system.
Solution 2:	Verify that the extraction folder is mounted to the SAP application server instance that the ABAP report is running on.
Solution 3:	Make sure to select an extraction folder with sufficient disk space. The team can assist you with determining the required disk space based on the output of the screening report.
Solution 4:	Do not mount/dismount the extraction folder in the file system while executing the transaction data report as this may cause errors in writing the output files

ISSUE: When validating the output of a transaction data report the team finds that not all of the required files have been created in the file system even though all reports completed without error.

Solution 1:	Some of the checkboxes have been deselected in the Table Groups section of the selection screen. Make sure no checkboxes are deselected (uncheck) any of the checkboxes unless advised by the team.
Solution 2:	An error occurred during the implementation of the ABAP reports causing the same ABAP code to be copied and pasted in two different reports. Make sure that the ABAP selection screen in each of the transaction data reports shows a different list of tables to be executed.

Solution 3:	Verify that only the current version of the transaction data reports is executed. Delete any older versions of the transaction data reports.	on data reports is ABAP transaction
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ISSUE: When validating the output of the transaction data report the output files do not contain all the relevant transaction data even though the reports are completed without error

Solution 1:	The ABAP transaction data reports can only be used to extract data from one SAP client at a time. Make sure that the reports are executed in the SAP client from which the data should be extracted. Make sure that this client is specified in the Client field in the selection screen.
Solution 2:	Verify that the parameters for Company Code and Fiscal Year match the customization of the SAP system. The company code must be the same as the customization. <u>Example:</u> Leading "0" numbers must be entered in the selection parameter if company code is "0100". Also, the Fiscal Year and the Date parameters must agree to the customizing of the fiscal periods. <u>Example:</u> If the SAP fiscal year is different from the calendar year, consider this when entering the Fiscal Year and Dates in the selection screen.

Transfer of the Data

ISSUE: High transfer times when copying the extracted data from the SAP application server to the encrypted hard drive.

Solution 1:	Copying the extracted data files from the SAP application in an uncompressed format may cause significant transfer time. If a compression program is available on the SAP application server, the extracted files should be compressed before copying them to the hard drive. The extracted files can be compressed to approximately 5% of their original file size using standard compression algorithms.
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ISSUE: The encrypted hard drive cannot be accessed or mounted after the data has been transferred

Solution 1:	Make sure that the encrypted partition on the external hard drive is not formatted when attaching the drive to a desktop computer
Solution 2	Make sure that the encrypted drive is properly un-mounted before the drive is removed from a desktop computer.

A.1 Reasoning for data download approach

The following is a brief summary of why [redacted] decided to design the ABAP reports and the different approaches taken to extract the data.

The extraction of data from the biggest transaction data tables in SAP is complicated and needs a table specific approach. The logic is built so that it can overcome potential threats such as:

- *Snapshot-to-Old issue*: Long running SELECT statement against the SAP Database Management System may time out if the activity in a particular SAP table is significant and the pointer of the selection loses its consistency (e.g. SAP Note 600141).
- *Full Table scan*: For certain SAP tables, the fields on which the [redacted] data requests can be limited, are not part of an index. To avoid extracting the whole [redacted] table and undergoing a full table scan, various approaches of incrementally looping through the tables instead of a single SELECT statement have been implemented.
- *Memory consumption*: To not overload the available memory (RAM) during the selection and to avoid affecting other background processes, only a limited amount of records per table is kept in memory at once.

The ABAP reports contain four different extraction scenarios as outlined in the next sections. The applicable extraction scenario for each table is determined based on the expected data volume and the available index structures on this table.

A.2 Normal extraction scenario

The following scenario describes the process for all the tables that are small in size such as the customizing tables like **T000** or **T001**.

Approach

A select statement fetches a set of records at a time from the database table and writes it to a data file internal table. It immediately writes it to a text file in a specified destination onto the application server.

Applicability

Transaction data tables with a small data volume or a low number of fields. All master data, dictionary and customizing tables.

Parameters in Selection Screen

None

Environmental and Performance Constraints

The **package size parameter** directly impacts the memory that is used for internal tables. An increase in the package size will increase the performance of the reports because a smaller number of loops are required. However, on systems with very restricted memory the default values should not be increased.

A.3 Number range loop scenario

Approach

- The number ranges for the primary key document number are put into a table.
- The report loops through each number range in partitions/packages and processes all the records through select statements by limiting on the necessary primary keys initially and later limiting the records separately on non-indexed fields individually by 'if' conditions.
- Sorted tables are used to increase efficiency. Once a number of header table records are processed, the corresponding footer tables such as **VBRP** or **CDPOS** are triggered. If activated, the records can be split into multiple packages and downloaded using the parallel download approach.
- The records are written into corresponding data files and then into text files to the specified destination onto the application server.

Applicability

Transaction data tables with large data volume that do not have an index that includes a relevant date field such as **VBAK**, **CDHDR**, and **RBKP**.

Parameters in Selection Screen

- **Package size** for the header tables which determines the number of posting documents that are included in one loop.
- **Parallel processing option** which is used to execute parallel downloads for records in the position tables (e.g. **VBRP**) using a remote function module (RFC). If this option is activated the package size for the position tables and the number of dialog processes for parallel downloads can be entered (up to 5).
- **Minimum/Maximum posting number** option which determines the lowest and the highest applicable posting document number for each number range interval. If this option is activated the lowest and highest number are selected from the transaction data tables and the values are then used as lower and higher boundaries for the number range loops to reduce the number of loops to be performed.
- **Use Number range intervals** option determines if the selection of the relevant number range intervals within a number range object (e.g. **RV_BELEG** for sales data) is restricted based on the customizing. This is relevant if a number range object is used for multiple tables (e.g. sales orders, deliveries and sales invoices). If this option is activated, only the relevant number range tables for a specific table are selected from customizing (e.g. **TVAK** for sales orders).

Environmental and Performance Constraints

- The package size parameter directly impacts the memory that is used for internal tables. An increase in the package size will increase the performance of the reports because a smaller number of loops are required. However, on systems with very restricted memory the default values should not be increased.
- Activating the **parallel processing option** will cause a temporary usage of up to five dialog processes in addition to the background process used by the job. Moreover, additional memory will be used for the execution of the function module. This needs to be considered on systems with very restricted memory and a small number of available dialog processes, especially when all three reports are executed in parallel.
- Activating the **Minimum/Maximum posting number** causes a select query on the transaction data table to determine the minimum/maximum posting document number for each number range object. This may cause a full table scan which can result in a *Snapshot-to-Old* issue. In this case, the Min/Max option has to be deactivated.

- If any entries in the customizing tables for the number range intervals (e.g. TVAK, TVLK and TVFK) have been updated or deleted in the period of the extraction, then you cannot use Number range intervals. Using the option may cause an incomplete extraction because some relevant number range intervals will not be included in the loops.

A.4 DaybyDay loop scenario

Approach

- If the Date field cannot be limited from a business requirement, all the dates from the entire database table are put into an internal table with the respective limitations on indexed fields. Otherwise, a list of dates from the database table within the selection screen date range is built into an internal table with the respective limitations on indexed fields.
- The report loops through each date and extracts all the records in packages by limiting on the necessary primary keys and the secondary indexed date field.
- Once a group of records are put into the internal table, corresponding footer tables are triggered such as the **FESEP** table.
- The records are written into corresponding data files and then into text files to the specified destination onto the application server.

Applicability

Transaction data tables with large data volume that have an index that includes a relevant date field (e.g. **BKPF**, **MKPF**).

Parameters in Selection Screen

- **Package size** for the header tables which determine the number of posting documents included in one loop.
- **Parallel processing option** is used to execute parallel downloads for records in the position tables (e.g. **BSEG**) using a remote function module. If this option is activated the package size for the position tables and the number of dialog processes for parallel downloads can be entered (up to 5).
- **Number of days in loop (#Days/DaybyDay loop)** which is used to determine the number of days that are part of one loop. In smaller systems this number may be increased to reduce the number of loops for a table.

Environmental and Performance Constraints

- The **package size parameter** directly impacts the memory that is used for internal tables. An increase in the package size will increase the performance of the reports because a smaller number of loops are required. However, on systems with very restricted memory the default values should not be increased. However, on systems with very restricted memory the default values should not be increased.
- Activating the **parallel processing option** will cause a temporary usage of up to five dialog processes in addition to the background process used by the job. Moreover, additional memory will be used for the execution of the function module. This needs to be considered on systems with very restricted memory and a small number of available dialog processes, especially when all three reports are executed in parallel.

A.5 Master data loop scenario

Approach

A structure with the primary keys (obtained from helper/lookup tables) is built.

2. These keys are used in packages to validate against the primary keys of the main tables that need to be extracted.
3. All records from the main tables are processed in packages for all entries in the previously triggered helper keys and written to an internal table at each step.
4. The records are written into corresponding data files and then into text files to the specified destination onto the application server.

Applicability

Transaction data tables that do not have an index that includes a relevant date field or a number range object attached to a table.

Parameters in Selection Screen

Transaction data tables that do not have an index that includes a relevant date field or a number range object attached to a table.

Environmental and Performance Constraints

The **package size parameter** directly impacts the memory that is used for internal tables. An increase in the package size increases the performance of the reports because a smaller number of loops are required. However, on systems with very restricted memory the default values should not be increased.

A.6 Authorization Objects

The User role authorization object details are in an excel file attached to this file. To view the Attachments panel click on the below:



User Role
Authorization Requi

The list below explains the authorizations the user needs to run the Data extraction program.

Authorization Object	Authorization Field and value	Programs which uses this authorization	Use of this authorization/Checks performed
S_GUI	ID 'ACTVT' FIELD '60'	Upload Program	This is used by the program to check if the user uploading the XML file (which has the scope and configuration data) has the authorization to load data into SAP system.
S_TABU_DIS	ID 'DICBERCLS' FIELD <table class> ID 'ACTVT' FIELD '03'.	Upload Program Config Program	Program checks if the user has authorization to display the contents of the tables belonging to the <table class>. ACTVT = '03' indicates display table entries.
S_TABU_NAM	ID 'TABLE' FIELD <table name> ID 'ACTVT' FIELD '03'.	Upload Program Config Program	Program checks if the user has authorization to display the contents of the <table name> specified. This check is performed if S_TABU_DIS check fails. ACTVT = '03' indicates display table entries.
S_DATASET	ID 'PROGRAM' FIELD PROGNAME ID 'ACTVT' FIELD '34' ID 'FILENAME' FIELD P_PATH.	Config Program Subroutine Program	Program checks if user has authorization to access operating system files. ACTVT = '34' indicates user can write and delete operating system files. File access can be restricted to a few known programs by using <PROGNAME>.
S_DATASET	ID 'PROGRAM' FIELD PROGNAME ID 'ACTVT' FIELD '33' ID 'FILENAME' FIELD P_PATH.	Download Program	Program checks if user has authorization to access operating system files. ACTVT = '33' indicates user can only read operating system files. File access can be restricted to a few known programs by using <PROGNAME>.
S_BTCH_ADM	ID 'BTCADMIN' FIELD 'Y'	Monitor Program Schedule Program	Checks if the user has authorization to perform all functions on all background jobs

Authorization Object	Authorization Field and value	Programs which uses this authorization	Use of this authorization/Checks performed
S_BTCH_JOB	ID 'JOBGROUP' FIELD " ID 'JOBACTION' FIELD 'PROT'	Monitor Program	In the monitor program, if the user does not have S_BTCH_ADM then this authorization is checked. This authorization object when assigned to a user, permits the user to perform different operations on select background jobs. 'PROT' lets the user display logs for other user's jobs
S_GUI	ID 'ACTVT' FIELD '61'	Download Program	This authorizes user to save lists displayed on SAP screen in local files (either unconverted, in spreadsheet format, or in Rich Text Format). This authorization can only be granted or denied for all lists in the system - you cannot restrict it to a selection of lists
S_TCODE	SA38 A/DATA_EX_20 SU53 AL11		User should be given access to the following transactions: SA38 – To run the Extraction program A/DATA_EX_20 ion is used to display the Extraction program's landing page SU53 – To check if the last authorization check for the user was successful AL11- To view the files on SAP Application server
S_SPO_DEV	LOCL %LOC		SAP names of the output devices, for which user is given authorization to print.
S_GUI	ID 'ACTVT' FIELD '2' ID 'ACTVT' FIELD '4'		User authorizations for activities 'Print' and 'Change' with reference to GUI objects
S_PROGRAM	ID ' P_ACTION' FIELD '*' ID 'GROUP' FIELD '*'		SAP checks if the user has authorization to execute ABAP programs by program group. By assigning this authorization, user can start the program, schedule the program as background job and maintain variants.

Authorization Object	Authorization Field and value	Programs which uses this authorization	Use of this authorization/Checks performed
S_WFAR_OBJ	ID OAARCHIV FIELD <Content Repository ID> ID OAOBJEKTE FIELD <Object Type> ID OADOKUMENT FIELD '*' ID ACTVT FIELD '03'	Attachment Extraction Program	Program checks if the user using the Attachment extractor program to extract documents from SAP Archivelinks, has the authorization to read information from them.
S_BDS_DS	ID CLASSNAME FIELD <Object Type> ID CLASSTYPE FIELD 'BO' ID ACTVT FIELD '30'	Attachment Extraction Program	Program checks if the user using the Attachment extractor program to extract documents from Generic Object Services or Business Document Services has the required authorization to retrieve that information

A.7 Document Extraction Tool

Original documents can be attached to an SAP Business object and stored in SAP via different methods as below:

- BDS(Business Document Services)
- GOS(Generic Object Services)
- AL(ArchiveLink)

The Generic Object Services or GOS offers functions that are available in various SAP applications. One of the commonly utilized function is the service to attach original documents or any relevant documents and stored in SAP, linked to the object. The object's link to the physical document in storage is in table **SRGBTBREL** with CAT_ID_A = 'BO'(Instances of BOR Object Types) and RELTYPE = 'ATTA'(Attachment).

The Business Document Services or BDS is an alternative to GOS and it offers about the same functions for the Business object as GOS. The object to the stored document link table is stored in one of the **BDS_CONN*** tables depending on the configuration of the Object type(or Class name in BDS) and only BOR object documents will be extracted (CLASSTYPE = 'BO').

ArchiveLink is a service integrated in the SAP Web Application Server for linking archived documents and the application documents entered in the SAP system. Documents are stored in an external storage system or content repository using SAP ArchiveLink. These archived documents can be displayed directly from a related application document. The object's link to the physical document in the content repository is in one of the **TOA*** tables (TOA01, TOA02, TOA03, TOAHR) depending on the configuration for the Object type.

Program A/DOC_EX_20 can handle the extraction of documents attached to SAP Business objects using the methods mentioned above. Use transaction code A/DOC_EX_20 to launch the program.

KPMG Data Extraction Download Attached Documents

SAP Business Object Selection

Object Type

SAP Object ID to

Content Repository ID

Constraints

Fiscal Year

Company Code

Period to

Document Number to

☐ Query DB table

--- Company Code Organizational Units ---

Sales Organization

Purchasing Org.

Valuation Area

Plant

Download Destination

☒ Server Directory Path

☐ Local Folder

Filename Prefix

Processing Options

Package Size

Maximum no. of hits

☐ Display Downloaded Objects in Report
☐ Joblog: Log ArchiveLink Object Fetch errors

Source

☒ Download from ArchiveLink

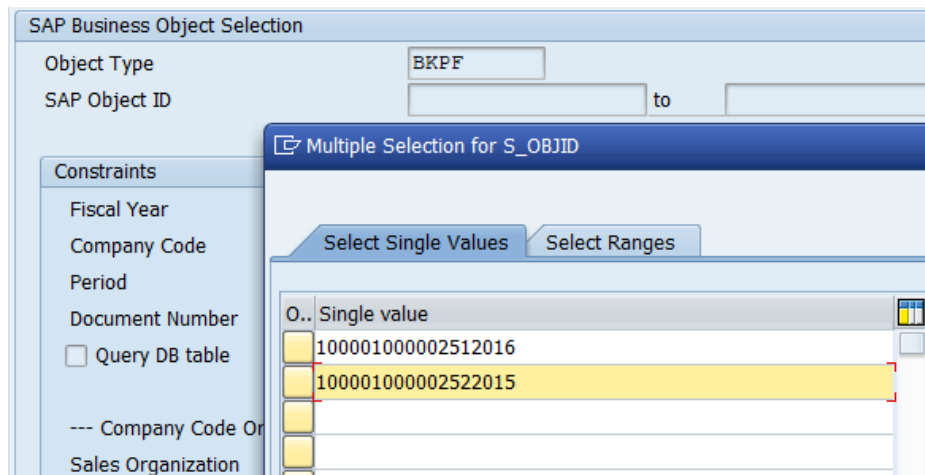
☐ Download from General Object Services(GOS)

☐ Download from Business Document Services(BDS)

☐ Download from all Content Repositories

There are 3 ways one can specify data selection:

- Enter 'SAP Object ID' directly. Any entries in the 'Constraints' are irrelevant here.
 - For example, for the Accounting document, enter 'BKPF' in Object Type and enter SAP Object ID's like '100001000002522016', where the first 4 is the Company code(BUKRS), next 10 is the Accounting document number(BELNR), then followed by the Fiscal year(GJAHR). Note that BUKRS, BELNR and GJAHR comprise the key fields of the SAP Object BKPF.



2. Unselect 'Query DB table' checkbox and enter selection values for 'Fiscal Year,' 'Company Code,' 'Document Number' whatever are the key fields of the 'Object Type'.
 - For example, for the object type BKPF, the key fields are Company code, Document number and Fiscal year. For object type VBRK and BUS2012, the Document number only.
 - The entries in 'Constraints' are used to populate entries in the 'SAP Object ID' prior to the data selection, but only when 'SAP Object ID' is empty.
3. Select 'Query DB table' checkbox and enter selection values for 'Fiscal Year,' 'Company Code,' 'Document Number', whatever are the key fields of the 'Object Type'. In addition, enter a date range in 'Period'.
 - This option also uses the entries in 'Constraints' to populate entries in the 'SAP Object ID' when it's empty except that it uses these entries to read the actual Database table. It fills the 'SAP Object ID' with document numbers that are actually existing in the database and for the specified period.

- Dates to be checked are hardcoded in the program for the following SAP Objects:
 - Posting date (BJPF-BUDAT) for BKPF
 - Create date (EKKO-AEDAT) for BUS2012
 - Billing date (VBRK-FKDAT) for VBRK and BUS2037
 - Document date (VBAK-AUDAT) for VBAK, BUS2032, BUS2034 and BUS2035
 - Delivery date (LIKP-LFDAT) for LIKP

Selection Screen Fields

Constraints

1. Checkbox: Query DB table
 - Option to select the document numbers from the database for the specified period
2. Fiscal Year
 - List of fiscal years to be considered for document extraction
3. Company Code
 - List of company codes to be considered for document extraction
4. Period
 - Date range to be considered for document extraction. Used only in conjunction with the selection of the 'Query DB table' option.
5. Document Number
 - enter list of actual document numbers or
 - enter number ranges (from and to) or
 - leave it blank and it will extract all documents for all combinations of Fiscal year and Company Code
6. Company Code Organizational Units: Sales Organization, Purchasing Org., Valuation Area and Plant.
 - With the 'Query DB table' option, data selection may be by Sales organization or Plant, etc. which can be derived directly from the current configuration of Enterprise assignments to the Company code. However, there are instances where the assignments during the period selected was different from the current configuration. These provides opportunity to override the current configuration. If no re-assignments occurred then just leave these empty and the current assignments will be defaulted.

SAP ArchiveLink Selection

1. Object Type
 - Only single value of SAP Object type can be entered
 2. SAP Object ID
 - TOA0*-OBJECT_ID values. These values are a concatenation of the key fields of SAP Object Type table.
 3. Content Repository ID
 - Identification of repository where the archived file is stored
- Storage Date
- Storage date for ArchiveLink

Download Destination

Enter server directory path where the files are to be downloaded or, optionally, if only a small number of documents are to be downloaded for testing, download directly to the frontend is also possible by choosing 'Local Folder'. Note that 'Local Folder' works only for Online execution.

2. Filename Prefix

- Option to prefix the output filenames

Processing options

1. Package size

- to manage the queries in order to avoid memory dumps
- SQL queries will be executed in batches of <package size> number of SAP Object ID's because too many could fail.

2. Maximum no. of hits

- works especially when Document number ranges are specified or no Document number is provided (indicating to extract for all documents)
- in order to avoid downloading large number of files, this is a means for setting a limit or sample size.

3. Display Downloaded Objects in Report

- All the objects downloaded are listed in the output report. This is an option to disable listing in instances where the list can be very long and could potentially cause a spool space problem.

4. Joblog: Log Archived Object Fetch errors

- When the SAP Archivelink referenced in the index table (TOA01, TOA02 or TOA03) could not be fetched from the Repository. This happens for example when the contents of the index tables are copied from the Production environment into the QA environment. Since the actual ArchiveLink repository is most likely not reachable from the QA environment, fetch error occurs and corresponding messages are logged in the job log.
- This provides an option to not display the error message because sometimes when there are too many messages in the job log, spool space problem can occur.

Execution

- Online execution should be used only when downloading only a few documents. For online execution, there is the option to download to the server directory or into a local desktop folder.
- For the most part, the program will be executed in the background (as a background job) where you can only download to the application server directory. This section describes how to execute the program in the background.

KPMG Archived Document Extract Program

SAP ArchiveLink Selection

Object Type: BKPF

SAP Object ID: to 

Content Repository ID: MA 

Storage Date: to 

Constraints

Fiscal Year: 2017 

Company Code: 1000 

Period: 01.01.2019 to 18.03.2019 

Document Number: to 

☒ Query DB table

--- Company Code Organizational Units ---

Sales Organization: 

Purchasing Org.: 

Valuation Area: 

Plant: 

Download Destination

☒ Server Directory Path: F:\usr\sap\

☐ Local Folder: C:\Temp\Test\

Filename Prefix: Z1_

Processing Options

Package Size: 9000

Maximum no. of hits:

☐ Display Downloaded Objects in Report ☐ Joblog: Log Archived Object Fetch errors

1. Fill the selection screen above. Go to Program => Execute in Background (or simply press F9)
2. Enter Output Device

Background Print Parameters

Output Device: LP01 

Number of copies: 1

Number of pages

☒ Print all

☐ Print from page: 0 To 0

Enter Start Time. Choose 'Immediate' to release the job right away. If scheduling for a later time, enter 'Date/Time'.

The 'Start Time' dialog box contains several input fields and buttons. At the top, there are five buttons: 'Immediate', 'Date/Time', 'After job', 'After event', and 'At operation mode'. Below these are four text input fields: 'Date/Time', 'After job', 'After event', and 'At operation mode'. At the bottom right, there is a 'Check' button with a green checkmark icon, and a close button with a red 'X' icon.

3. Click the **SAVE** icon after entering Start Time

The 'Start Time' dialog box is shown with the 'Immediate' button selected. The 'Date/Time' field now contains the text 'Immediate start'. The 'After job', 'After event', and 'At operation mode' fields remain empty. A 'Periodic job' checkbox is located at the bottom left, which is currently unchecked. The 'Check' button at the bottom right is circled in blue.

4. Then you should see the message that the background job was scheduled.

The 'Processing Options' dialog box displays several settings: 'Package Size' is set to 9000, 'Maximum no. of hits' is empty, 'Display Downloaded Objects in Report' is checked, and 'Joblog: Log Archived Object Fetch errors' is checked. At the bottom, a green checkmark icon is next to the message: 'Background job was scheduled for program /KPMGDA/ARCHIVELINK_EXTRACT_91'.

Execute transaction code SM37 to monitor/check the background job. When job status is 'Finished', select the job by checking the checkbox then click the Spool pushbutton to see the output report or click the Job log pushbutton to see the contents of the Job log for messages.

Job Overview										
Refresh Release Spool Job log Step Application servers										
Job overview from: 09.04.2019 at: : : to: 09.04.2019 at: : : Selected job names: * Selected user names: ITAFFMDEV ☐ Scheduled ☒ Released ☒ Ready ☒ Active ☒ Finished ☒ Canceled ☐ Event controlled Event ID: ☐ ABAP program Program name :										
JobName	Spool	Job doc	Job CreatedB	Status	Start date	Start time	Duration(sec.)	Delay (sec.)	Sched. start date	Sched. start time
☒ /KPMGDA/ARCHIVELINK_EXTRACT_91			ITAFFMDEV	Finished	09.04.2019	18:05:37	0	0	09.04.2019	18:05:37
☒ /KPMGDA/ARCHIVELINK_EXTRACT_91			ITAFFMDEV	Finished	09.04.2019	21:07:18	2	0	09.04.2019	21:07:18
*Summary							2	0		

- To see the output files, execute transaction code AL11 and navigate to the directory specified in the 'Server Directory path'.

Assumptions and Dependencies

- Time and space needed for stored original document extraction is going to be considerable as documents are stored in external storage and time taken for reading those documents is going to be significant.
- Any Object type can be downloaded except when the 'Query DB table' checkbox is selected. In this version of the prototype the constraints used to query the database are hardcoded in the program and should work only for object types BKPF, VBRK, BUS2012, BUS2017, BUS2032, BUS2034, BUS2035, BUS2037, BUS2081 and LIKP.
 - BKPF (Accounting Document - BKPF):
BUKRS IN S_COCODE And
BUDAT IN S_DATE And
BELNR IN S_DOCNBR
 - BUS2012 (Purchase Order - EKKO):
BUKRS IN S_COCODE And
AEDAT IN S_DATE And
EBELN IN S_DOCNBR
 - BUS2081 (Incoming Invoice - RBKP):
GJAHR IN S_YEAR And
BUKRS IN CO_CODE And
BUDAT IN S_DATE And
BELNR IN S_DOCNBR
 - VBRK/BUS2037 (Customer Billing Document - VBRK):
BUKRS IN CO_CODE And
FKDAT IN S_DATE And
VBELN IN S_DOCNBR
 - VBAK/BUS2032/BUS2034/BUS2035 (Sales Document - VBAK):
VKORG IN IVKORG And
AUDAT IN S_DATE And
VBELN IN S_DOCNBR
 - LIKP (Outbound Delivery - LIKP):
VKORG IN IVKORG And
LFDAT IN S_DATE And
VBELN IN S_DOCNBR
 - BUS2017 (Goods movement - MKPF):
BUDAT IN S_DATE And
MBLNR IN S_DOCNBR

In the future if there other Object types that need to be downloaded using DB query, the program needs to be modified to insert the necessary constraints.

A.8 File Size Estimation Button

File size estimation feature added to 2020 extraction tool helps estimate table record counts and sizes for the scope selected in the XML file. This information is valuable to calculate total expected size of the export directory used by the raw data files before the start of extraction. Errors related to export directory running out of space can now be avoided and a better plan can be made to extract data in batches.

KPMG Data Extraction Tool

Upload

Config

Scheduling

Monitoring

Download

Metadata Profile

STANDARD

Constraints

Client

800

Fiscal Year

2007

Company Code

1000

Objectclasses

ANLA

Period

01.01.2007

to

31.12.2007

Language

EN

--- Company Code Organizational Units ---

Sales Organization

Purchasing Org.

Valuation Area

Plant

Destination

Export Directory

.\

Filename Prefix

Clean Directory Before Extraction

☐

Estimate File Size

File Compression

Compress files

☒

Max. file size (MB)

500

After each Table

☒

After each File partition

☐

Delete files after Compress?

☒

Compress files in parallel

☒

the 'Estimate File Size' button to start the estimation, which is carried by the program A/DATA_EX_SIZE_EST_20. This program runs in background mode and writes the unt and calculated size for each table to the spool (SP01) and the job log (SM37).

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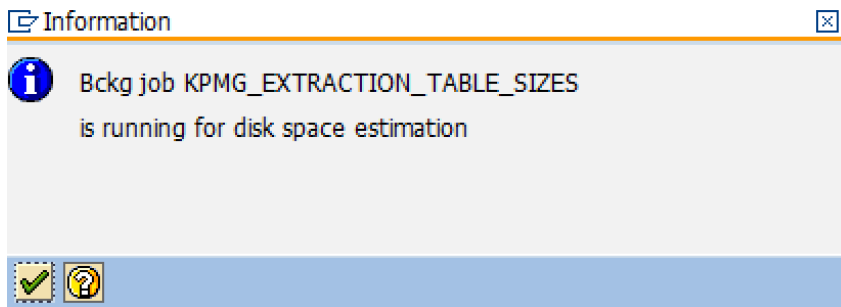
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Use SM37 to check the status of the job **EXTRACTION_TABLE_SIZES**.

SAP Job Overview

Job overview from: 22.05.2020 at: : :
to: 22.05.2020 at: : :
Selected job names: *
Selected user names: TVARUGHESE

☐ Scheduled ☒ Released ☒ Ready ☒ Active ☒ Finished ☒ Canceled
Event-Driven Event ID:
ABAP program Program name :

JobName	Spool	Job doc	Job Created	Status	Start date	Start Time	Duration(sec.)	Delay (sec.)
☒ KPMG_EXTRACTION_TABLE_SIZES			TVARUGHESE	Finished	22.05.2020	07:23:47	323	0
#Summary							323	0

Output of the program can be verified by clicking on the 'Spool' button.

SAP Graphical display of spool request 315508 in system DC6

Data statistics Number of
Records passed 248

KPMG Data Extraction Tool - File Size Estimator Report
Date: 22.05.2020

Report	Group	Object ID	Table Name	H Table Name	Num of Rows	Row Size	File Size
SYS	Customizing Tables	ADR6	ADR6		323.437	290	100589026
SYS	User-Roles/Profiles	AGR_AGRS	AGR_AGRS		3.839	63	264928
SYS	User-Roles/Profiles	AGR_DEFINE	AGR_DEFINE		7.257	103	878199
SYS	User-Roles/Profiles	AGR_PROF	AGR_PROF		3.102	106	366093
SYS	User-Roles/Profiles	AGR_TCODES	AGR_TCODES		208.802	83	19209827
SYS	User-Roles/Profiles	AGR_TEXTS	AGR_TEXTS		46.461	119	6086444
SYS	User-Roles/Profiles	AGR_USERS	AGR_USERS		30.061	77	3036280
FI	FI Master data: Accounts	AN EK	AN EK		2	402	1630
FI	FI Master data: Accounts	AN EP	AN EP	AN EK	24	135	8832
FI	FI Master data: Accounts	AN KT	AN KT		125	183	25586
FI	FI Master data: Accounts	AN LA	AN LA		1.323	448	819572
FI	FI Master data: Accounts	AN LB	AN LB		9.886	222	3440800
FI	FI Master data: Accounts	AN LC	AN LC		11.786	447	6824588
FI	FI Master data: Accounts	AN LP	AN LP		9.827	537	6781218
FI	FI Master data: Accounts	AN LT	AN LT		0	132	169
SYS	Customize:other tables	AP QI	AP QI		291	191	67103
FI	Accounting Documents	BK PF	BK PF		30.037	684	28746443
FI	Accounting Documents	AC CTIT	AC CTIT	BK PF	1.691	136	783404
FI	Accounting Documents	BSEC	BSEC	BK PF	0	597	756
FI	Accounting Documents	BSEG	BSEG	BK PF	46.321	1.210	88706118

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